Linking Human Reconstruction Models for Policy Summarizations to Task Structure

Bachelor thesis by Jaron Schäfer
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1. Review: Prof. Dr. Constantin Rothkopf
2. Review: M.Sc. Inga Ibs
Darmstadt



TECHNISCHE UNIVERSITÄT DARMSTADT

Human Sciences Department

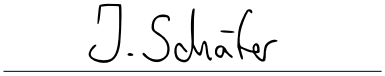
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Jaron Schäfer

Abstract

As AI agents continue to integrate into our daily lives, establishing effective collaboration between machines and humans becomes crucial. This can be accomplished by enabling algorithms to provide explanations for their behavior. A promising approach in this domain is model-based policy summarization, where subsets of actions that characterize agent behavior are extracted to explain their conduct. However, as highlighted by previous research, ensuring accurate reconstruction of an agent's policy by humans requires careful model selection. In this study, we present a simple navigation task and employ imitation learning and inverse reinforcement learning user models to extract policy summarizations explaining the behavior of an optimal agent. Our primary objective is to investigate how different task structures influence the reconstruction models employed by humans. Additionally, we explore the impact of transferring summary information between tasks on the extent to which models accurately represent human behavior, and on human choice of reconstruction models. Through a computational experiment, we analyze and compare user models concerning their capacity to extract and reconstruct summaries within our navigation task. The obtained results challenge prior findings and reveal considerable effects of task structure on reconstruction accuracy. In a complementary subject study, we assess the human ability to solve a navigation task using policy summarizations while examining the influence of task structure on the selection of reconstruction models. Our research indicates that determining the reconstruction model employed by humans remains ambiguous, but we find evidence of a connection between model choice and task structure. Remarkably, humans exhibit proficient use of policy summarizations as guidance. However, it becomes apparent that the chosen models may not entirely capture human behavior, suggesting potential areas for further improvement in this domain.

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1. Introduction

In an ever-advancing world where artificial intelligence (AI) continues to grow, it becomes crucial to establish optimal means for interacting with this technology. While the primary challenge in recent years has centered around instructing computers, algorithms, and robots, the landscape has evolved. We now find ourselves at a stage where enabling algorithms to instruct, teach, and explain to humans is of equal importance. With AI agents (systems capable of performing human-like tasks) becoming increasingly integrated into our lives, it is necessary to bridge the gap between machines and humans, facilitating humans to collaborate effectively with algorithms. This includes not only providing instructions but also imparting knowledge, teaching new concepts, and explaining their actions and decision-making processes (Gohel et al., 2021; Xu et al., 2019). For instance, the efficiency of AI systems that assist in navigating or teaching can be significantly enhanced when the robot is capable of effectively communicating its operational guidelines (Amershi et al., 2019).

Given that humans naturally acquire and understand knowledge and behavior through demonstrations, one prevalent approach for explaining agent policies to humans is through the use of demonstrations. However, the effectiveness of demonstrations relies on their ability to provide informative and insightful sets of examples. Consequently, researchers have focused on summarizing an agent's overall behavior by showcasing its actions across various states. This approach, as demonstrated by Amir et al. (2019), aims to optimize knowledge transfer by emphasizing the significance of well-executed demonstrations in enhancing instructional effectiveness.

The problem of policy summarization or extracting subsets of state-action pairs that globally characterize the agent's behavior can be approached in different ways. A common approach used in research assumes a computational model of how humans will generalize from the summaries provided. These models are then used to optimize the summary for reconstructing the agent's full policy (Huang et al., 2017).

In the context of learning from demonstrations, there are two primary classes of models: Imitation Learning (IL) and Inverse Reinforcement Learning (IRL) (Dixon et al., 2020). IL refers to a learning approach where individuals replicate others' actions to achieve desired outcomes. It assumes that humans directly mimic and reproduce the demonstrated behavior without necessarily inferring the underlying intentions or goals of the demonstrator. IRL on the other hand involves inferring the underlying reward function or goals from observed behavior. It assumes that humans build models of others' behavior by trying to understand the intentions, preferences, or rewards that drive the observed actions.

Following the model approach, Lage et al. (2019) used an IL-based and IRL-based model to explore their utility for agent policy summarization. In a computational study, they showed that a mismatch between the model used for summary extraction and the model used for policy reconstruction leads to poor reconstruction accuracy. In a human-subject study, they additionally observed that individuals employ different models for reconstructing policies in different contexts and further demonstrated that aligning the summary extraction model with the specific reconstruction context can enhance performance. These findings imply that understanding in which task contexts people deploy which model can assist in matching the extraction to the reconstruction model. Ultimately, this could lead to improved human performance.

Considering these results, we extend the research conducted by Lage et al. (2019) with four main objectives. Firstly, we seek to validate that aligning the extraction and reconstruction models is the most effective strategy for generating accurate reconstructions. Additionally, we aim to explore the impact of task structure on

human and model behavior. By ”task structure,” we refer to the fundamental organization and arrangement of elements and interactions within a given task. In particular, we are interested in exploring how different task structures influence the selection of reconstruction strategies used by humans. Lastly, we intend to investigate the predictive abilities of the models regarding human performance and behavior.
For this purpose, we designed a navigation task that involves utilizing a policy summary to determine an optimal path. This navigation task is further divided into two conditions: the ”Base-Condition” and the ”Transfer-Condition”. In the Base-Condition, a corresponding summary is provided, which can directly be used to solve the given condition or reconstruct the policy. Conversely, in the Transfer-Condition, only a related but not corresponding summary is given, requiring the transfer of information to effectively solve the condition. With this setup, we can investigate how transferring summary information between conditions impacts our objectives, making the findings more relevant to real-life situations. If the Transfer-Condition leads to different performances of the models, this would even allow identifying the model that best captures human conduct. To potentially influence the human selection of a reconstruction model, we introduced three variations to the task structure. Incorporating these variations we conducted both a computational experiment, where models were tasked with extracting and reconstructing summaries, and a subject experiment, where humans were challenged to solve our task using these summaries. By combining these approaches, we sought to gain a comprehensive understanding of how models and humans utilize policy summaries, allowing us to address our main questions.
In this thesis, we present our research, starting with an overview of the methodological foundations for our navigation task and introducing the models used for policy summarization (chapter 2). Following that, we will detail our computational experiment (chapter 3) and present the corresponding results (chapter 4). Further, we will explain our subject experiment, including evaluation strategies and supplementary information (chapter 5). The results of the subject experiment will be reported in chapter 6. Finally, we will engage in a thorough discussion of our findings (chapter 7) and offer directions for future research (chapter 8).

2. Methods

In this chapter, we present a comprehensive overview of the methodological framework that underpins our work. We explain the mathematical aspects related to the task environment, the determination of optimal agent behavior, and the choice and implementation of task structure variations (section 2.1). Furthermore, the chapter defines the process of model-based policy summarization (section 2.2), introduces the reconstruction models (section 2.3), and outlines the extraction models (section 2.4) utilized in the study.

2.1. Task Environment

We model our task environment as an instance of a deterministic finite-state Markov Decision Process (MDP). In the following, we will explain the mathematical properties of an MDP and the exact representation of our task.

2.1.1. Markov Decision Process

An MDP is a mathematical framework used to model decision-making (White III & White, 1989). An environment is said to be Markovian when the probability of transitioning from one state to another depends only on the current state and action, and not on the past history of states and actions. This is known as the Markov property (Puterman, 1990).
Formally, our MDP is defined by a tuple (S, A, T, R, γ), with the following properties:

- S is a finite set of states.
- A is a finite set of actions.
- $T(s, a, s')$ is the transition function. It specifies the next state s' that an agent will transition to when taking action a in state s. In our deterministic MDP, this is represented as a one-hot vector, indicating the unique state being transitioned to.[1]
- $R(s, a, s')$ is the reward function and denotes the reward received when transitioning from state s to state s' when taking action a. We defined the reward function as:

$$R(s, a, s') = \varphi(s')^T \cdot w^* \tag{2.1}$$

  where $\varphi(s')$ is the one-hot state dimension vector corresponding to state s' and w^* is the reward weight vector.
- γ is a discount factor ($0 \leq \gamma \leq 1$), which determines the importance of future rewards relative to immediate rewards.

[1] A one-hot vector is a binary vector where all elements are zero except for a single element, which is set to one.

In general, the goal of an agent in an MDP is, to find a policy $\pi : S_i \rightarrow A$ that maximizes the expected sum of discounted rewards obtained while navigating through the MDP. Hence, there also exists an optimal policy $\pi^* : S_i \rightarrow A$ within the environment that assigns the best action for each state to maximize rewards in the long run. We used a value iteration algorithm to obtain this policy.

2.1.2. Value Iteration

The optimal policy is computed using a value iteration algorithm. This algorithm iteratively updates the value function using the Bellman optimality equation until convergence (Dai et al., 2011). Updates are given by:

$$V_{k+1}(s) = \max_{a \in A} \sum_{s' \in S} P(s, a, s')[R(s, a, s') + \gamma V_k(s')], \quad \forall s \in S, \tag{2.2}$$

where $V_k(s)$ is the estimated value function at iteration k. Termination of the algorithm occurs when the difference between consecutive value functions falls below a specific threshold.
The optimal policy can then be obtained by greedily selecting the action that maximizes the sum of the immediate reward and the discounted value of the next state, using the final estimated value function:

$$\pi^*(s) = \arg\max_{a \in A} \sum_{s' \in S} P(s, a, s')[R(s, a, s')\gamma V^*(s')], \quad \forall s \in S \tag{2.3}$$

Using this, π^* is guaranteed to be the optimal policy, i.e., the policy that achieves the maximum expected sum of discounted rewards over all possible policies.

2.1.3. Gridworld

Our research focuses on a Markov Decision Process that serves as a navigation task example. The MDP is represented by a 7x5 gridworld, consisting of a maximum of nine state dimensions [2]. It consists of different states, including "regular states" and "goal states". Regular states, in our case, are the states within the gridworld that incur a step reward of -1 for each movement. On the other hand, the goal state is a specific state in the gridworld that the agent aims to reach. Upon reaching the goal state, the agent receives a reward of 100. These goal states, being terminal, were implemented as absorption states, which means that the agent remains in that state indefinitely without any further actions or transitions. Note that each gridworld contained one or two goal states. To represent the rewards for each state dimension we utilize reward weights in the form of $w^* = (-1, 100, c, d, e, f, g, h, i)$, where $c, d, e, f, g, h, i \in [-1, -20]$, depending on the specific task variation.
We emphasize that the policy is determined through value iteration, as outlined in the previous section (2.1.2). To account for a deterministic scenario, we employ a discount factor of $\gamma = 1$. An illustration of an exemplary gridworld is shown in figure 2.1. The available actions for an agent within our gridworld are limited to moving one state up, down, left, or right. Furthermore, each gridworld configuration includes a set of trajectories denoted as B, comprising individual trajectories starting from each possible initial state and following the optimal policy to reach a goal state. Consequently, an agent's or player's primary objective, starting from an initial state s_0, is to navigate through the gridworld towards a goal state while maximizing the cumulative reward along the chosen path.

[2] State dimensions in a Markov Decision Process refer to the properties or attributes that define a specific state within the environment, including the reward assigned to that state.

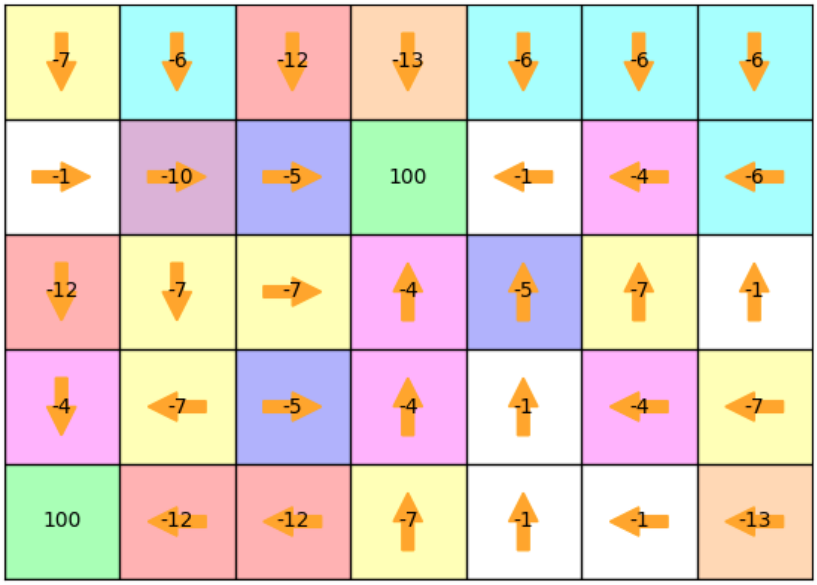


Figure 2.1.: Example of a possible gridworld, with green tiles representing the goal states and white tiles as regular states. Arrows indicate the optimal action for each state, and numbers represent the reward for each state. The reward weights are $w^* = (-1, 100, -7, -6, -12, -13, -4, -5, -10)$. Note that this is an illustrative example and does not reflect the actual appearance of the gridworlds used in the experiment.

2.1.4. Task

As mentioned in the introduction our navigation task consists of a "Base-Condition" and a "Transfer-Condition". These Conditions can be considered as separate tasks represented by individual gridworlds. In each condition, the task is to navigate from a starting state to a specific goal state. The primary objective is to minimize the negative rewards encountered during this navigation. In order to identify states that should be avoided and to determine appropriate behavior within the gridworld, a summary is provided for each condition. This summary includes five trajectories selected from the optimal trajectory set B. These trajectories serve as illustrative examples of paths that should be taken into consideration in order to successfully accomplish the task. The Base-Condition and Transfer-Condition share similar underlying properties, such as reward structures and task variations, but they differ in their specific gridworld layouts. In the Base-Condition, a corresponding summary is provided, specifically tailored to this exact condition. In the Transfer-Condition, only the information from the summary of the Base-Condition is provided, making it necessary to transfer the information to solve the condition effectively. An illustration depicting two exemplary conditions that were used in the subject study is presented in figure 5.1.

Task Structure Variations

Given the lack of direct information in the existing literature regarding the influence of task structures on human reconstruction model choice, we introduce three variations that will be considered in our study. Lage et al. (2019) hypothesized that IL is commonly used in less familiar domains. Also, research on human explanations reveals a tendency for people to favor simpler explanations with fewer causes (Lombrozo, 2016). Consequently, when confronted with a more complex summary, individuals might naturally tend to lean towards utilizing IL, similar to their approach when dealing with unfamiliar domains. This could be a mechanism

to manage cognitive overload and enhance decision-making efficiency, as IL offers a computationally more efficient approach. Therefore we expect that a more complex task environment, leading to a more intricate summary, will impact human learning toward IL. To investigate these hypotheses, we have selected variations based on task density, the number of state dimensions, and the number of distraction dimensions.

Density Imagine a scenario in which a robot guides workers through a densely packed storage facility, offering concise instructions for efficient navigation and item location. In such an environment, the density of obstacles and narrow aisles will influence the choice of reconstruction method. As complexity increases and more detailed summarizations become necessary to cope with the overwhelming nature of the environment, humans could be likely to rather deploy IL than IRL, benefiting from its cognitive efficiency. In our task, density refers to the proportion of states where dimensions differ from the regular-state dimension. To explore the influence of density, we consider two variations:

- High Density (HD): In a high-density gridworld, we sampled the proportion of states that differ from the regular-state dimension using a distribution: $X \sim \mathcal{N}(0.75, 0.05)$. On average, 26 out of 35 states have a state dimension that differs from the regular-state dimension.
- Low Density (LD): In a low-density gridworld, we sampled the proportion of states that differ from the regular-state dimension using a distribution: $X \sim \mathcal{N}(0.4, 0.05)$. On average, 14 out of 35 states have a state dimension that differs from the regular-state dimension.

Number of State Dimensions In various contexts, state dimensions refer to the number of variables or attributes necessary to fully describe the state or configuration of a system or entity. Consider the context of a finance management algorithm that provides policy summarizations to aid individuals in personal finance management. Here, the number of dimensions involved could significantly impact the selection of the reconstruction method. As the number of dimensions increases, applying IRL becomes more challenging. Thus, individuals are more inclined to opt for IL for higher-dimensional tasks, given its greater cognitive efficiency. In our task, the number of state dimensions refers to the count of different state dimensions present in the gridworld. We consider two variations:

- High number of state dimensions (HF): In this variation, we randomly assign 7, 8, or 9 different state dimensions with probabilities $p(0.25, 0.5, 0.25)$.
- Low number of state dimensions (LF): In this variation, we randomly assign 4, 5, or 6 different state dimensions with probabilities $p(0.25, 0.5, 0.25)$.

Number of Distraction Dimensions Williams et al. (2010) found that explanations given to people can have a negative impact if they don't help learners find important patterns or if they accidentally confuse them with distractions. In light of this, we aimed to investigate how distraction dimensions influence the policy summarization process and impact human behavior. We define distraction dimensions as state dimensions that appear to be relevant but do not influence the system's behavior more than the regular dimension. The presence of distraction dimensions can affect the choice of reconstruction methods used by humans, diverting their attention away from meaningful dimensions and introducing unnecessary complexity to the task. Therefore, we want to investigate how distraction dimensions influence the policy summarization process and how they affect human reconstruction behavior. In our task, a distraction dimension to a state dimension with the same reward as the regular-state dimension (step reward -1) but has a different representation.

We distinguish between two variations:

- High number of distraction dimensions (HS): In this variation, the probability of a feature being a distraction dimension is $P(D) = 0.75$.
- Low number of distraction dimensions (LS): In this variation, the probability of a dimension being a distraction dimension is $P(D) = 0$.

These task structure variations allow us to examine how different configurations of density, number of state dimensions, and number of distraction dimensions affect the reconstruction methods used by humans in inferring policy summaries within the MDP. An example of a gridworld with a high density, a high number of state dimensions and a low number of distraction dimensions (HD_HF_LS) is depicted in figure 2.1.

2.2. Summarizing Agent Policies

Policy summarization refers to the problem of summarizing an agent's policy into concise and informative example states and actions that should be conveyed to help a user gain a comprehensive understanding of an agent's policy. The goal is to provide users with a high-level understanding of the agent's behavior or skill without overwhelming them with excessive details. Policy summarization aims to distill the essential information necessary for comprehension and decision-making (Amir et al., 2019). This problem of extracting the states that globally characterize an agent's policy is usually approached through two methods.
The first approach integrates heuristic techniques that utilize agent Q-values and the similarity of states to identify a range of varied and significant states (Amir & Amir, 2018; Huang et al., 2018). Sometimes also inconsistencies between the policies of distinct agents are taken into account (Amitai & Amir, 2022).
The second approach follows the machine teaching framework formalized by (Zhu et al., 2018). In this approach, a learning model is assumed, which should represent the students' learning behavior. The objective of machine teaching is to select the minimal set of teaching examples, or demonstrations, that guide the learner towards a specific target model, such as a policy. The model is used to test the learning effect obtained by the examples and therefore helps to optimize the summary referring to the learning model. Our work extends the second approach using two models that represent different methods of human extrapolation.
Human Extrapolation refers to the cognitive process by which individuals utilize their existing knowledge, patterns, or experiences to make predictions or inferences about situations that extend beyond the scope of their current knowledge or direct observation. This describes the process humans subconsciously employ when using policy summarizations to understand agent behavior. Investigating human learning from a reinforcement learning perspective, research reveals two primary planning or extraploration strategies employed by humans: model-based and model-free systems (Daw et al., 2005; Gläscher et al., 2010). In a model-based system, humans acquire a cognitive model of the environment, which enables them to compute rewards and transitions. This cognitive model serves as a mental representation of the world and allows individuals to simulate different paths and scenarios. By mentally simulating the consequences of various actions and considering the associated rewards, individuals can evaluate the potential outcomes and make informed decisions. Complementing the model-based system is the model-free system, which relies on previous experiences and learned associations between states and actions. Unlike the model-based system, the model-free system does not build an explicit cognitive model of the environment. Instead, it learns action values directly based on observed actions. Through accumulated knowledge, the model-free system estimates the value of different actions without explicitly representing the underlying concepts and rewards. Gläscher et al. (2010) emphasizes the distinction between these two systems, highlighting the model-based system's reliance on a cognitive model and the model-free system's dependence on learned associations from past experiences.

However, it is important to note that human learning will mostly involve a combination of these strategies, depending on the task, individual preferences, and available information.
Considering the machine teaching approach, these findings can be used to develop two models of human extrapolation. One learning model used in the literature is the inverse reinforcement learning model (IRL) by (Huang et al., 2018), which recovers the agent's reward function from demonstrations and refers to the model-based system. In their paper "Exploring Computational User Models for Agent Policy Summarization", Lage et al. (2019) proposed another approach based on Imitation Learning (IL) referring to the model-free system.

For our study, we choose to adopt the model-based approach to policy summarization, utilizing the IRL and IL model presented by Lage et al. (2019). However, while they focused on finding a minimal set of state-action pairs, we formalize the problem as follows:
The goal of summary extraction is to identify a minimal set of trajectories $T = t_1, t_2, ..., t_k$ of size k, where each trajectory consists of l_n state-action pairs $t_n = (s_1, a_1), ..., (s_{ln}, a_{ln})$, that maximizes the similarity ρ between the agent's policy π^* and the policy $\hat{\pi}$ induced by T under a specific computational model M. In other words, we seek the examples that best represent the agent's policy in a compact manner, while taking into account a reconstruction model. Mathematically, this can be expressed as:

$$\max_{T \in B, |T|=k} \rho(\hat{\pi}(T, M), \pi^*) \tag{2.4}$$

where B is the set of all trajectories starting from each state and following the agent's true policy π^*, and $\hat{\pi}(T, M)$ is the approximate policy obtained by applying the computational model M to the summary T. ρ is a measure of reconstruction accuracy (refer section 2.5), which represents the similarity between the agent's policy π^* and the reconstructed policy $\hat{\pi}$.

2.3. Models of Human Extrapolation

To model human extrapolation we use the Maximum Entropy (Max-Ent) algorithm by Ziebart et al. (2008) for an IRL-based reconstruction and the Gaussian Random Field Model (GRF) by (Zhu, Lafferty, et al., 2003) for an IL-based reconstruction.

2.3.1. IRL - Maximum Entropy

Inverse reinforcement learning is the problem of inferring the reward function of an agent, given its policy or observed behavior (Arora & Doshi, 2021; Ng, Russell, et al., 2000). In the context of policy summarization, this is accomplished by analyzing a collection of expert demonstrations, denoted as $T = t_1, t_2, ..., t_k$. Each trajectory t_i in T consists of a sequence of state-action pairs, represented as $(s_1, a_1), ..., (s_l, a_l)$, where the expert navigates through the state-action space. In our specific scenario, the expert trajectories are acquired by following the optimal policy of our task.
Using the model-based approach to IRL of (Ziebart et al., 2008) the challenge of learning a policy from observed trajectories is done by optimizing a linear function. To facilitate state computations, a feature function $\phi : S \to \mathbb{R}^d$ that maps each state to a d-dimensional vector is utilized. In Maximum Entropy Inverse Reinforcement Learning (MaxEnt-IRL), the goal is to match the expected frequency of feature visitation in expert demonstrations with the expected frequency of feature visitation by an agent following the policy obtained using the recovered rewards. This ensures that the features characterizing the states of the environment are visited with a similar frequency by both the expert and the agent.

Feature expectations, denoted as $\mu_{\pi}^{(s,a)}$, represent the average frequency with which each feature is visited by the expert. They are computed as follows:

$$\mu_{\pi}^{(s,a)} = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t \phi(s_t), |, \pi, s_0 = s, a_0 = a\right] \tag{2.5}$$

where $\mu_{\pi}^{(s,a)}$ represents the feature expectations resulting from starting at state s, taking action a, and following the policy π thereafter. Here, $\phi(s_t)$ represents the feature vector of state s_t, and γ is the discount factor that determines the importance of future rewards.
MaxEnt-IRL combines the maximum entropy principle with an iterative refinement process to learn a policy and reward function from observed demonstrations. By maximizing the entropy of the learned policy while aligning the feature visitation frequencies, MaxEnt-IRL ensures exploration and accurate inference of the underlying reward function. The iterative process begins with initial reward weights and iteratively minimizes the discrepancy between the expected feature counts of the learned policy and the observed feature counts. With each iteration, the reward weights are adjusted to better align the feature visitation frequencies, gradually capturing the expert's behavior. This iterative process continues until convergence or until a specified amount of epochs is reached (Ziebart et al., 2008).

2.3.2. IL - Gaussian Random Field

Imitation learning is a technique where the goal is to learn a policy function that imitates an expert policy based on a set of observed demonstrations. Let $T = t_1, t_2, ..., t_k$ be a set of expert demonstrations, where each trajectory t_i consists of state-action pairs $t_i = (s_1, a_1), ..., (s_l, a_l)$. The learning algorithm aims to estimate a policy function $\hat{\pi}$ that maps states to actions by minimizing the difference between the distribution of actions under the learned policy and the expert policy. This can be formulated as an optimization problem of the form:

$$\min \sum_{(s,a)\in\mathcal{D}} \mathcal{L}(a, \pi^*(s)),$$

where $\mathcal{L}$ represents a suitable loss function that measures the discrepancy between the predicted action and the expert action.
While inverse reinforcement learning aims to learn an underlying reward function that explains the expert's behavior, imitation learning aims to learn a policy directly from expert demonstrations by mapping states to actions. In IL, the focus is on mimicking the expert's actions without explicitly considering the underlying motivation behind them. Similar to Lage et al. (2019) we chose the Gaussian Random Field approach proposed by (Zhu, Lafferty, et al., 2003) for the imitation learning reconstruction. The GRF model leverages graph structures to represent data points and their relationships effectively. In this model, data points (states) are depicted as vertices in a graph, connected by weighted edges (actions) that reflect their similarity.
To measure the similarity between pairs of data points, a kernel function denoted as $v(\phi(s), \phi(s'))$ is employed. Here, v acts as the kernel function, and ϕ represents the feature function. The feature function maps each state s to a corresponding feature set, transforming it into a feature vector that captures its distinctive characteristics. The kernel function v evaluates the similarity between two feature sets $\phi(s)$ and $\phi(s')$, projecting them into a higher-dimensional feature space. In the binary classification setting[3], the GRF

[3] In a binary classification setting, we have a task where the goal is to classify data points into one of two classes or categories. Each data point is associated with a set of features, and based on these features, we want to determine whether it belongs to the positive class (class 1) or the negative class (class 0).

model makes predictions by propagating labels, or actions, through the graph. The action probabilities are determined by the following equation:

$$p(D) = \frac{1}{Z_\beta} \exp\left(-\beta \sum_{s,a \in D} \sum_{s',a' \in D} \frac{v(\phi(s), \phi(s'))(a - a')^2}{2}\right), \tag{2.6}$$

where D represents the set of labeled data points, a and a' are the labels of data points s and s', respectively, and β is a tunable inverse temperature parameter that controls the randomness (set as $\beta = 1$). The normalization factor Z_β ensures that the action probabilities sum to one over all possible actions.
To extend the GRF model to the multiclass classification setting, a one-vs-rest classification strategy is employed, where separate binary classifiers are trained for each class Zhu, Ghahramani, et al. (2003). In this setting, the goal is to predict the class of each input data point, and each classifier distinguishes instances of the class of interest from instances of all other classes.
To make predictions in the multiclass setting, the Laplacian matrix of the graph is computed. The Laplacian matrix captures the graph structure and facilitates label smoothing and regularization. The combinatorial Laplacian matrix L is defined as $L = \text{diag}(\sum_{s' \in D} v(\phi(s), \phi(s'))) - V$, where D represents the set of labeled data points, ϕ is a kernel function mapping data points to a feature space, and V is the matrix of similarity values for all pairs of data points. After computing the Laplacian matrix, the GRF model uses it to predict labels for the unlabeled data points. The prediction is obtained by solving the equation:

$$\hat{\pi}U = -L_{UU}^{-1} L_{UT} \pi_T^*, \tag{2.7}$$

where $\hat{\pi}U$ is a vector containing the predicted labels for the unlabeled data points, L_{UU}^{-1} is a submatrix of the Laplacian matrix representing the connections between the unlabeled data points, L_{UT} is a submatrix representing the connections between the labeled and unlabeled data points, and π_T^* is a vector containing the labels for the labeled data points. The resulting predicted labels $\hat{\pi}$ can be further binarized by thresholding at 0.5 to obtain prediction probabilities.
It should be noted that the Laplacian matrix is used to smooth the labels across the graph and produce coherent results. The Laplacian operator helps to propagate information between neighboring nodes in the graph so that nodes with similar features are likely to have similar predictions. The Laplacian operator can also regularize the predictions and prevent overfitting (Zhu, Ghahramani, et al., 2003).

2.4. Summary Extraction

For the summary extraction, the goal is to find the set of trajectories T that maximizes the reconstruction quality for the two models for human summary reconstruction. This involves maximizing the similarity p between π^* and $\hat{\pi}$.

2.4.1. IRL - Machine Teaching

Considering an IRL extraction we aim to find the minimal set of trajectories required for our IRL model to recreate a reward function that leads to an optimal policy. This reconstructed reward function should fall within the behavioral equivalence class (BEC) of π^*. To achieve this, we employ the SCOT machine teaching algorithm proposed by Brown and Niekum (2019) which involves extracting trajectories that should maximize the quality Max-Ent reconstruction (Zhu, Ghahramani, et al., 2003).

Behavioral Equivalence Class

The set of reward functions that lead to an optimal policy π^* is known as the behavioral equivalence class (BEC) of π^*. That means that there is usually not only one optimal reward function but a space of functions that all lead to an optimal policy.
The BEC of π^* (all reward functions that lead to the same policy as π^*) can be expressed by the intersection of halfspaces given by the following constraints:

$$w^T \left(\mu_{\pi^*}^{(s,a)} - \mu_{\pi^*}^{(s,a')} \right) \geq 0, \forall \langle s, a \rangle \in D, \forall a' \in A \tag{2.8}$$

Here, D is the set of all state-action pairs, A is the set of actions available to the agent, $w \in \mathbb{R}^k$ are the reward weights, and $\mu_{\pi^*}^{(s,a)}$ are the expected feature counts for an optimal behavior starting at state s taking action a. These can be defined as:

$$\mu_\pi(s,a) = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t \phi_{s_t} | \pi, s_0 = s, a_0 = a] \quad (2) \tag{2.9}$$

Where $\mu_\pi(s,a)$ are the feature expectations count resulting from starting at state s , taking action a and following the policy π thereafter. ϕ_{s_t} is the feature vector of state s_t and γ is the discount factor.

Set Cover Optimal Teaching

The SCOT machine teaching algorithm, proposed by Brown and Niekum (2019), selects trajectories that provide the most informative data for the Max-Ent algorithm. This is achieved by translating all possible trajectories into a corresponding set of halfspace constraints. Redundant constraints are then eliminated using linear programming. The algorithm proceeds in a greedy manner, iteratively searching for trajectories that cover the largest number of uncovered constraints. This iterative process continues until all constraints are covered or a desired summary length is reached, resulting in the identification of the minimal set of trajectories. By extracting this minimal set, the resulting reward function lies within the behavioral equivalence class (BEC) of π^*. For a more detailed explanation of the SCOT algorithm and BEC, please refer to the work of Brown and Niekum (2019).

2.4.2. IL - Active Learning

Similar to the concept of IRL extraction, our objective in IL extraction is to identify a minimal set of trajectories in B that maximizes the similarity p between π^* and $\hat{\pi}$ but for IL reconstruction.
To achieve this, we employ an active learning algorithm based on the work of Zhu, Lafferty, et al. (2003), but with modifications to accommodate the knowledge of the optimal policy as proposed by Lage et al. (2019).
The algorithm follows an error reduction strategy, where it greedily selects the trajectory that minimizes the 0/1 loss on all unseen states. That means: Given a trajectory, all states in that trajectory are considered to be labeled, whereas the rest is unlabeled. The Algorithm then searches for the trajectory, which leads to the best prediction for the unlabeled states given the GRF model. This can be represented by the following equation:

$$t^* = \arg\min_{t \in T} \sum_{s \in S : (s,a) \notin t} \mathcal{L}(\pi^*(s), \hat{\pi}(s)) \tag{2.10}$$

where $\mathcal{L}(\pi^*(s), \hat{\pi}(s))$ is the 0/1 loss between the truth action $\pi^*(s)$ and its predicted action $\hat{\pi}(s)$ given the GRF model. $s \in S : (s,a) \notin t$ describes all states that are not included in a trajectory and therefore unlabeled.

To select the trajectory(s) that minimize the loss, the algorithm uses a greedy approach. It iteratively selects the trajectory that leads to the best prediction for the leftover unseen states, given the already selected trajectories. This process continues until the desired summary length is reached.

2.5. Measures

In this section, we establish and outline the measures that will serve as fundamental evaluation tools throughout this thesis. These measures are carefully chosen to assess and quantify various aspects of our study. By defining these metrics upfront, we ensure a robust and systematic approach to analyzing the performance, effectiveness, and similarities involved in the context of our research.

Reconstruction Accuracy

Reconstruction accuracy (ρ) serves as a metric to assess the proficiency of models in reconstructing a policy from a summary. In alignment with (Lage et al., 2019), we define reconstruction accuracy as the precision of predictions made on states that are absent from the summary. Essentially, it measures the proportion of states in the reconstructed policy that are excluded from the summary but accurately labeled by the model.

Performance

Performance refers to the effectiveness or success in accomplishing the task (MDP). It is measured as the achieved reward compared to the highest attainable reward within the particular MDP context. That means the percentage of the optimal reward. By expressing the attained reward as a percentage of the maximum possible reward, this metric facilitates an evaluation of the model or subject's performance in the given MDP scenario and allows for meaningful comparisons between the trials.

Summary Similarity

Summary similarity refers to the proportion of states utilized to solve an MDP that are also presented in the summary. This measure indicates the extent to which states from the summary were employed while solving the task, shedding light on whether humans utilized IL rather than IRL techniques.

Path Similarity

Path similarity is defined as the degree of resemblance between two taken paths. This measure enables a comparison between a subject's or model's path and the optimal path to assess their similarity. It is quantified as the proportion of states shared by both paths, revealing how closely they align.

3. Computational Experiment

Using the models of human extrapolation we conducted computational experiments, for both Base- and Transfer-Conditions. Our aim was to understand how the task structure affects reconstruction quality and to replicate and validate the findings of Lage et al. (2019). To do so, we tested the four different combinations of extraction and reconstruction models (IL-IL, IL-IRL, IRL-IRL, IRL-IL) to see if matching the model used for summary extraction with the human-reconstructed model affects the quality of results. Further, we investigated how the models are impacted by the different variations in our task environment.
For the policy summaries, we used a fixed size of five trajectories per summary. Consequently, the extraction algorithms were tasked with selecting five trajectories from the set B of trajectories generated for each gridworld which provided the most information for the matching reconstruction model. Note that the trajectories can have different lengths, resulting in different total amounts of states presented in the summaries.
We generated 50 sets of conditions for each variation combination (HD_HF_HS, HD_HF_LS, HD_LF_HS, HD_LF_LS, LD_HF_HS, LD_HF_LS, LD_LF_HS, LD_LF_LS), consisting of the reward weights, a Base-Condition gridworld and Transfer-Condition gridworld. Be aware that Base- and Transfer-Conditions had the same underlying rewards and the same variational properties but a different layout. In total, we had 400 condition sets, which were tested with every extraction and reconstruction method combination resulting in a total of 3200 accuracy results.

3.1. Reconstruction Accuracy

In order to evaluate the quality of reconstruction, we employed the reconstruction accuracy metric as outlined in section 2.5. This involved generating a summary for a specific condition and providing it to the model, which then attempted to reconstruct the underlying policy. Distinct methodological approaches were utilized for the Base- and Transfer-Conditions.

Base-Condition

Our Base-Condition experiment aimed to replicate and expand upon the computational experiment conducted by Lage et al. (2019) in the gridworld domain. We made several adjustments to the experimental setup, with a primary focus on evaluating the reconstruction accuracy of different model combinations. By assessing the effectiveness and performance of various models in accurately reconstructing the underlying structure of the environments, we sought to gain insights into their capabilities and compare our results with those of Lage et al. (2019). The reconstruction accuracy metric served as a key indicator of how well the models captured the essence of the original environments. We anticipated that using matching extraction and reconstruction models would yield higher accuracy, aligning with the findings of Lage et al. (2019).
To achieve the best possible accuracy while considering our specified summary size, we optimized the hyperparameters of each model for every single condition. This hyperparameter optimization process allowed us to fine-tune the models and maximize their reconstruction quality. Additionally, we employed different feature functions for each model to align them with the underlying hypothesis. By exploring various hyperparameter

configurations and feature sets, we aimed to unlock the full potential of each model and obtain comprehensive insights into their performance in accurately reconstructing the underlying environment structure.

GRF For the Gaussian Random Field algorithm, we used various hyperparameters, including the kernel type (ϕ), the length scale, and the degree for the polynomial kernel, to optimize the algorithm. We specifically utilized different kernel types, such as the Radial Basis Function (RBF) and polynomial kernels. Regarding the length scale, we varied the values between 0.1 and 1.0 for both the RBF and polynomial kernels. Furthermore, we explored different degrees for the polynomial kernel, including 2 and 3. It is worth mentioning that although we initially considered a broader range of values, in alignment with Lage et al. (2019), we determined that these specific options resulted in the highest accuracy.
For the IL reconstruction, we employed a feature function ϕ that considered the surrounding states. Our goal was to capture meaningful information about each state by taking into account its immediate neighbors. The resulting feature vector was created by combining the state dimension vectors of the focal state and its nearby states. To focus on the direct influences and relationships between states, we only included the four neighboring states, since diagonal actions are not possible anyway.

Max-Ent Concerning the Max-Ent algorithm, we only varied the learning rate between 0.5 and 1. The maximum epochs of the algorithm was kept fixed at 300 and the termination threshold at 1^{-5}. For the feature set, we exclusively utilized the single state dimension vector for each state. This feature set configuration resulted in a reconstructed reward weight vector, which not only captured the essential information but also aligned with the hypothesis of inverse reinforcement learning. It is noteworthy that our choice of feature set deviated from the approach employed by Lage et al. (2019) in their study.

Transfer-Condition

Also in the Transfer-Condition, our objective was to evaluate the quality of reconstruction. Specifically, we examined how effectively the models reconstructed the underlying policy when provided with a summary from the corresponding Base-Condition. By measuring the reconstruction accuracy in the Transfer-Condition, we aimed to assess the models' capacity to generalize their learned knowledge and apply it to novel environments. This investigation builds upon the previous research conducted by Lage et al. (2019) and contributes valuable insights into determining the more suitable approach for knowledge transfer.
In the case of IL reconstruction, regardless of whether the summary was generated through IL or IRL, we employed the states within the summary as labeled states for the GRF algorithm. Simultaneously, we considered all other states within the gridworld as unlabeled states. Therefore the GRF algorithm could use the actions in the labeled states to predict the actions in the unlabeled states.
Regarding the IRL reconstruction, regardless of whether the summary originated from IRL or IL, we computed the reward weights for the gridworld corresponding to the summary. These reward weights were subsequently utilized in the Transfer-Condition, where we calculated the reconstructed policy using value iteration.

3.2. Model Performance

An additional intriguing aspect of the models is how they perform in the task of our subject study. Details on the whole study and the explicit task and trial creation are presented in chapter 5. The model performance can be calculated by applying the measure defined in section 2.5. This involves using the reconstructed policies for each condition and solving the condition by following this policy starting at the corresponding start state. The reward obtained following this path is then used to calculate the performance.
It is important to note that when starting from a start state and following the reconstructed policy, two scenarios can occur. In the first case, the policy successfully guides the agent to a goal state. However, in certain instances, the agent may enter a loop, unable to reach the goal state. In such cases, the reward obtained is -inf, resulting in a performance score of 0. These performance results serve a dual purpose. Firstly, they allow us to compare the performance of the participants in our subject study to that of our models. Secondly, we can compare the trajectories taken by the participants to those predicted by the trajectory model toward the goal state. This comparison will provide valuable insights into the effectiveness and accuracy of our models in replicating human behavior. Results of the model performance are included in section 6.3.

3.3. Computational Experiment Modifications

Given that we replicated and extended the computational experiment conducted by Lage et al. (2019), it is crucial to clarify the differences and adjustments we made. By providing this explanation, we can better understand any potential discrepancies in results and gather insights for future research.
Firstly, while they used three different domains, we focused on a single one that is comparable to their random gridworld domain but with a navigation task included. Therefore the overall task does differ in some aspects.
Another significant difference lies in the summarization process. Lage et al. (2019) used summaries with a limited number of states, selecting summary trajectories based on their length to avoid exceeding a maximum state count. They explored summary sizes of [12, 24, 36, 48, 60] state-action pairs with a trajectory length of [1] for the IL model and [1, 2, 3, 4] for the IRL model. In the gridworld domain, they determined the summary size for the reconstruction accuracy measurement to ensure the best-performing IRL methods remained consistent even with an increased summary size. In contrast, our approach focused solely on the number of trajectories, which we set to five for both models. We believed that including trajectories leading entirely to the goal states, rather than showing only a part of the way, was more realistic for a navigation task. Consequently, the number of states in our summaries varied, even though all had the same number of trajectories. Furthermore, we did not adjust the summary size to maximize reconstruction quality. Instead, we kept it fixed because doing so would artificially increase reconstruction accuracy without being realistic for human usage.
In terms of algorithms, we used different features for the Max-Ent methods compared to their approach. We selected features to realistically match the human decision-making process when using these methods. For the Max-Ent algorithm, we only used state dimensions since including neighbor states' information didn't contribute to informative policy reconstruction. However, for the GRF algorithm and extraction algorithms, our feature sets were similar to theirs. Lastly, we tailored our rewards to match the characteristics of our navigation task, resulting in differences from Lage et al. (2019). This adaptation was essential to ensure that our models learned relevant and task-specific behaviors, aligned with the nature of navigation.

4. Results - Computational Experiment

In this chapter, we present the results of our computational experiment. We examined four model combinations to reconstruct the policy in our task environment. For this, we utilized 400 different gridworlds, incorporating our chosen variations, and optimized the extraction and reconstruction algorithms to achieve the highest reconstruction accuracy for each gridworld and model combination.

In general, we observed that IL summaries tended to have fewer states and shorter trajectories compared to IRL summaries, even though they were chosen from the same set. Also, we discovered the importance of optimizing the accuracy of each reconstruction. Without optimizing the reconstruction accuracy was substantially lower. This indicates that the models are not very robust, and one needs to find the best parameters for each specific gridworld to get accurate results. We also found the GRF algorithm to be much more runtime efficient than the Max-Ent algorithm. Finally, we noticed that the Max-Ent algorithm sometimes gave slightly different results with the same settings, while the GRF algorithm consistently produced the same reconstruction accuracy every time. Using these further insights the computational experiment was adjusted many times to achieve the results presented in the following.

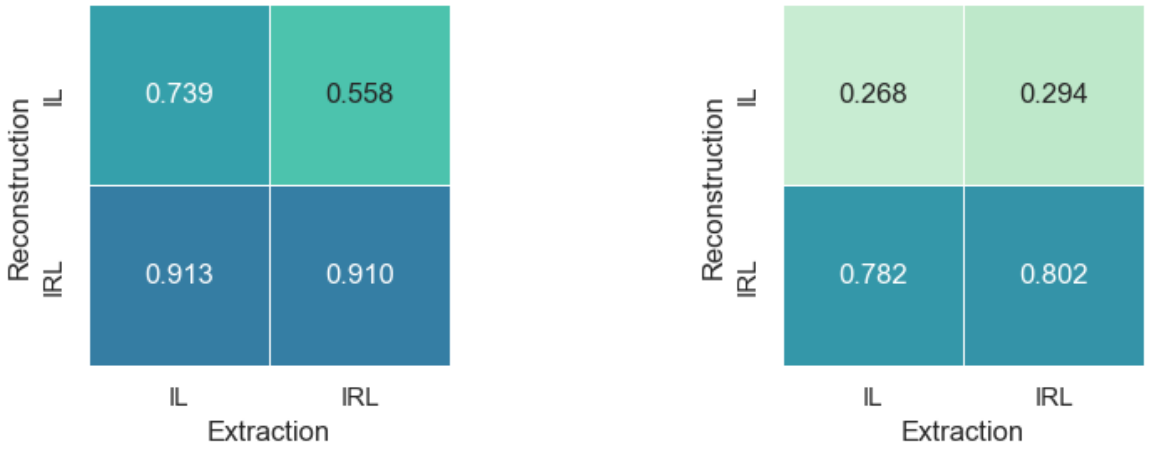
As mentioned we calculated the reconstruction accuracy for different model combinations. We also distinguished between Base- and Transfer-Conditions and later between the task structure variations. Figure 4.1 shows the mean reconstruction accuracy for each model combination per condition.

For the Base-Condition (figure 4.1a), the results indicate that, overall, the IRL reconstruction outperforms the IL reconstruction model in terms of reconstruction accuracy. Interestingly, in certain instances, the IL summary achieves slightly higher accuracy than the IRL summary. While the IL reconstruction method generally exhibits lower accuracy, there is a noticeable distinction between using an IL summary and an IRL summary. Matching the summary and extraction method in the IL reconstruction model does lead to increased accuracy; however, this improvement does not seem to apply to the IRL reconstruction model.

Figure 4.1b illustrates the results of the Transfer-Condition, revealing a clear loss of accuracy compared to Figure 4.1a. In this condition, the reconstruction accuracy of the IL model is remarkably low, almost at chance levels (0.25). Surprisingly, with an IRL summary, the accuracy is slightly higher for this reconstruction. Nevertheless, the IRL reconstruction remains relatively good and still surpasses the IL-IL accuracy of the Base-Condition. However, only a minor difference between IL and IRL summaries can be observed in this context, consistent with the results from the Base-Condition.

Further, we investigated the influence of the chosen task variation on reconstruction accuracy, also distinguishing between the four extraction-reconstruction model combinations. Figure 4.2 shows these results of the Base-Condition as the distribution of reconstruction accuracy results for each variation per model combination.

The subplots indeed suggest that the variations do influence the reconstruction accuracy. In each model combination, the median of the boxplots appears to differ across one variation type, with some models being more affected than others. Overall, there seems to be a tendency that higher density leads to slightly worse accuracy, and a similar pattern is observed for the number of state dimensions. However, the impact of the number of distraction dimensions on accuracy is not consistent for all model combinations. In the case of the IL reconstruction, a higher number of distraction dimensions increases accuracy. Conversely, for the



(a) Base-Condition

(b) Transfer-Condition

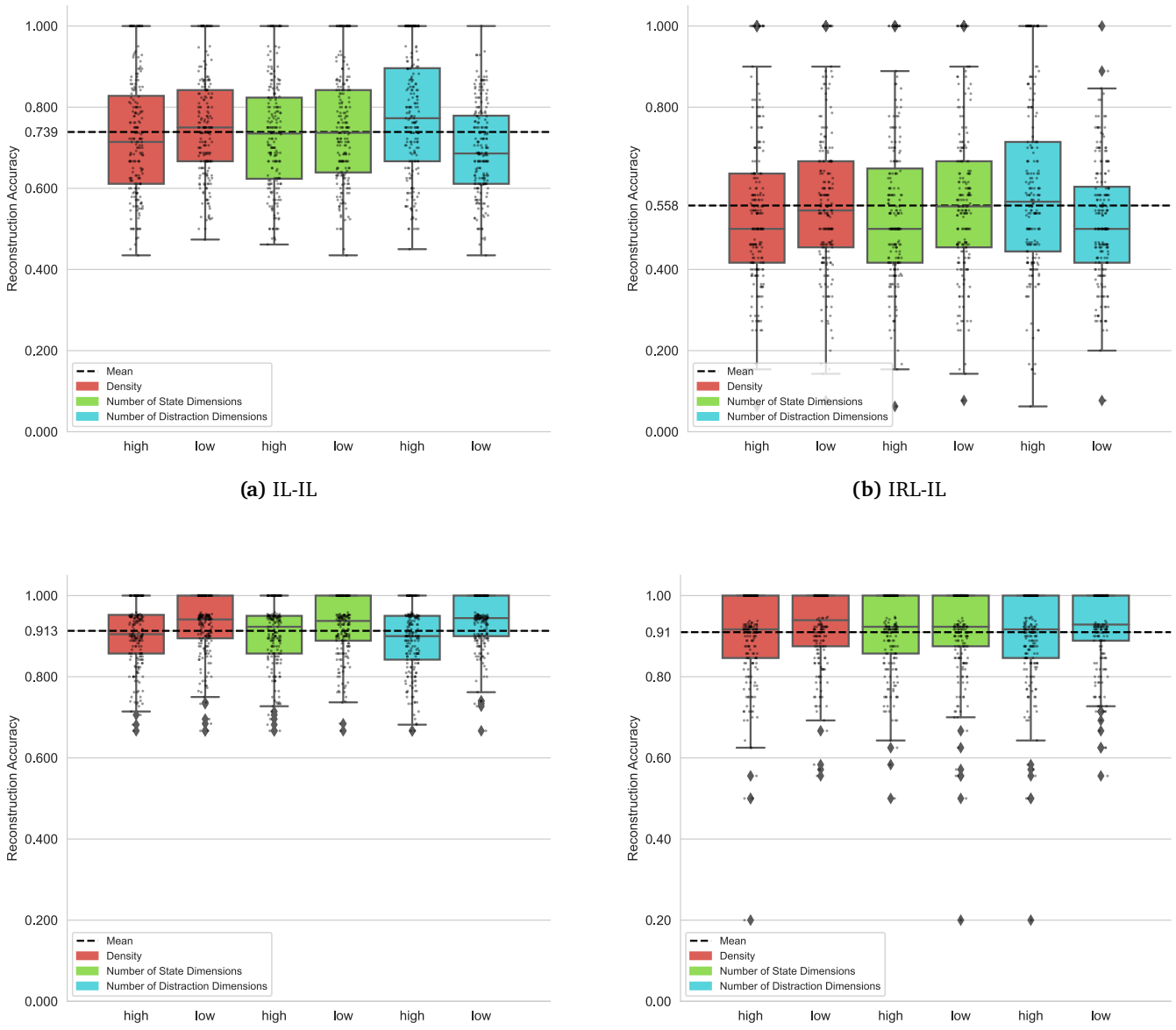
Figure 4.1.: Mean reconstruction accuracy per model combination and condition. The x-axis shows the model type used for summary extraction and the y-axis the model used for reconstruction. Each tile represents the mean of 400 optimized reconstruction accuracies computed with the specified model combination.

IRL reconstruction, the opposite is observed, where a lower number of distraction dimensions leads to better accuracy. It is important to note that these differences are relatively small and could potentially be influenced by various factors.

The most intriguing aspect lies in the comparison between the plots of one reconstruction model. By comparing figure 4.2a and figure 4.2b, we observe that they appear to be influenced to a similar extent, despite having drastically different means. In the case of IRL reconstruction, a contrasting situation arises when comparing figure 4.2c and figure 4.2d. Although they have nearly the same mean, the distribution of reconstruction accuracies is influenced to different extents. While the IRL-IRL combination seems to be influenced only slightly, the boxplots for the IL-IRL combination display strong differences. Visually, not only do the medians differ between the high and low variations within each combination, but the entire boxes also exhibit noticeable shifts.

The results for the Transfer-Condition (figure A.3) are provided in Appendix A. These results demonstrate similar differences with one exception. Notably, they clearly illustrate that the number of distraction dimensions has different impacts on the reconstruction models, just as we investigated in the Base-Condition. For the IL reconstruction model, a high number of distraction dimensions appears to lead to similar or overall higher reconstruction accuracy. On the contrary, for the IRL reconstruction model, the opposite is true. It becomes evident that a low number of distraction dimensions leads to better accuracies.

We further wanted to investigate the influence of task structure variations between Base- and Transfer-Conditions for all model combinations. In Figure 4.3, we present the distribution of reconstruction accuracy for each variation and condition. The right side of each violin plot illustrates the distribution for the Transfer-Condition, while the left side depicts the Base-Condition. The dotted lines within the violin plots represent the quartiles and the median. Upon analyzing the violin shapes, we find that there are no significant differences in accuracy between the variations within each condition. However, when comparing the Base- and Transfer-Conditions overall (as displayed in figure 4.1), a clear distinction emerges in the general reconstruction performance. This comprehensive plot sums up accuracy across all model combinations and consistently reveals that the reconstruction accuracy for the Transfer-Condition is lower than that of the Base-Condition.



(a) IL-IL

(b) IRL-IL

(c) IL-IRL

(d) IRL-IRL

Figure 4.2.: Distribution of the reconstruction accuracy for the Base-Condition. Each subplot illustrates a different model combination. The colors of the boxplots represent the variation types further distinguished between high and low. Black scatter points show each data point per boxplot. The dotted line in each of the four subplots shows the mean across all variations.

While no notable accuracy differences are apparent between all variations, we do observe subtle fluctuations in shape and median between the high and low manipulation within one variation, particularly concerning the number of distraction dimensions. Specifically, for the Transfer-Condition, a high number of distraction dimensions yields fewer low reconstruction accuracies but also fewer high ones. This finding aligns with the differences shown in figure A.3 figure. Additionally, in the Base-Condition a higher number of distraction dimensions leads to a higher median (median=0.85, IQR=0.67,0.94) compared to the low variation (median=0.82, IQR=0.61,0.94). These differences indicate more lower accuracy reconstructions, aligning with the earlier observations. Whereas we initially hypothesized that the various conditions would be similarly impacted by variations in task structure, it appears that there is a discrepancy when it comes to density. In the Base-Condition, a lower density yielded higher overall accuracy (high: median=0.82, IQR=0.62,0.93, low: median=0.85, IQR=0.67,0.95), whereas in the Transfer-Condition, a higher variation in density led to improved accuracy (high: median=0.6, IQR=0.26,0.8; low: median=0.53, IQR=0.26,0.81). Consequently, this plot shows that the Base- and Transfer-Conditions are affected by variations in distinct manners.

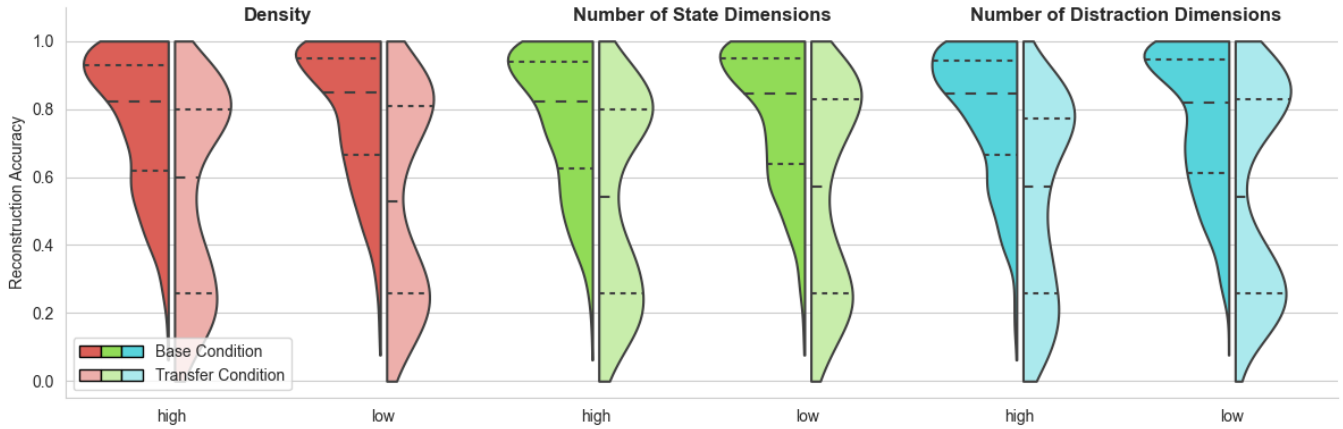


Figure 4.3.: Comparison of reconstruction accuracies between the Base and Transfer-Conditions for all variations. Each violin plot displays the distribution of reconstruction accuracies for the Base-Condition (left side) and the Transfer-Condition (right side). The dotted lines inside each violin represent the median and quartiles.

5. Subject Experiment

Using the task environment described in section 2.1, we conducted a subject study to explore the influence of task structure variations on behavior, performance, and the choice of reconstruction model. The following chapter will provide an explanation of our subject experiment, beginning with the basic structure in section 5.1. We will then delve into the reasons for this specific layout, how we plan to evaluate the study, and how the individual trials were created and selected based on the results of the computational experiment. Lastly, we will provide some methodological and participant details.

5.1. Experiment Design

The experiment comprised 4 training trials and 32 normal trials. Among the normal trials, each trial was repeated twice, once with an IRL summary and once with an IL summary. To ensure trials were not recognized, we applied horizontal and vertical mirroring to the IL trials. Thus, we had 16 trials with an IRL summary and 16 with an IL summary, which were shown in a random order for each subject. During each trial, participants encountered two distinct gridworlds: the Base-Condition and the Transfer-Condition. Both gridworlds shared the same reward weights and structural variables, but they differed in layout. The main objective for participants was to identify the optimal path leading to a specific goal state in each gridworld, utilizing the corresponding summary. Importantly, in the Transfer-Condition (the second gridworld), participants were only given the summary from the corresponding Base-Condition.

5.1.1. Experimental Task Environment

To translate the objective of our navigation task to an experiment we used the analogy of a robot that has to be controlled on a foreign planet. We illustrated the different state dimensions of the gridworlds as various types of landscapes, distinguished by their unique colors or textures. For example, if there were four dark blue areas in a particular gridworld, they all represented the same type of landscape and therefore the same state dimension. Each landscape had a size of > 2 visual degrees. The appearance of an example trial is shown in figure 5.1. The main goal for the participants was reaching a specific goal, which was associated with a particular landscape type, the goal-state dimension. We used the analogy of minimizing fuel consumption to include the rewards and the path costs. We explained that different landscape types would result in varying levels of fuel consumption, meaning the reward for each state dimension. Thus, the participants must carefully evaluate their choices and select the most optimal path to efficiently conserve fuel. Furthermore, we informed the participants that gray landscapes, which served as the regular-state dimension, consistently required the lowest fuel consumption on every planet. However, it was also possible for other landscape types to consume fuel at the same rate as the gray landscapes, representing distraction dimensions in the environment. Note that the appearance of each landscape/state dimension changed every trial randomly, only the regular-state dimension states or the grey landscapes were included in every trial and had the same optic. Each trial presented the participants with two different environments on the same planet (Base-Condition and the Transfer-Condition). At the beginning of each condition, the participants started with a full tank of fuel or a reward of 0. While the fuel consumption for each landscape type remained consistent across both

conditions, the arrangement of individual landscapes did vary. To include the summaries we told participants their command center presented them with example paths that demonstrated the optimal routes from various starting points to the goal. These were displayed at the top of the screen (see 5.1). The arrows within the paths indicated the direction of the optimal route. Each summary trajectory surface had a size of approx. 5 degrees of visual angle. We instructed participants to use these example paths to identify the goal landscape type and to gain insights into which landscape type consumed a significant amount of fuel. Participants were encouraged to utilize this knowledge when planning their own paths but had also the opportunity to use these trajectories in their path. It was important for participants to understand that the command center had limited information about the planet and that the second environment they would encounter had not been explored yet. As a result, they would only receive example paths from the first environment as guidance. This transfers to the idea of a Transfer-Condition where only the summary of a gridworld with similar reward weights is known. We emphasized that there was no time pressure, allowing participants to take their time and carefully consider each of their steps. We also encouraged them to make informed decisions and guide the robot strategically using arrow keys on their keyboard.

5.1.2. Methods

Eye tracker and Hardware

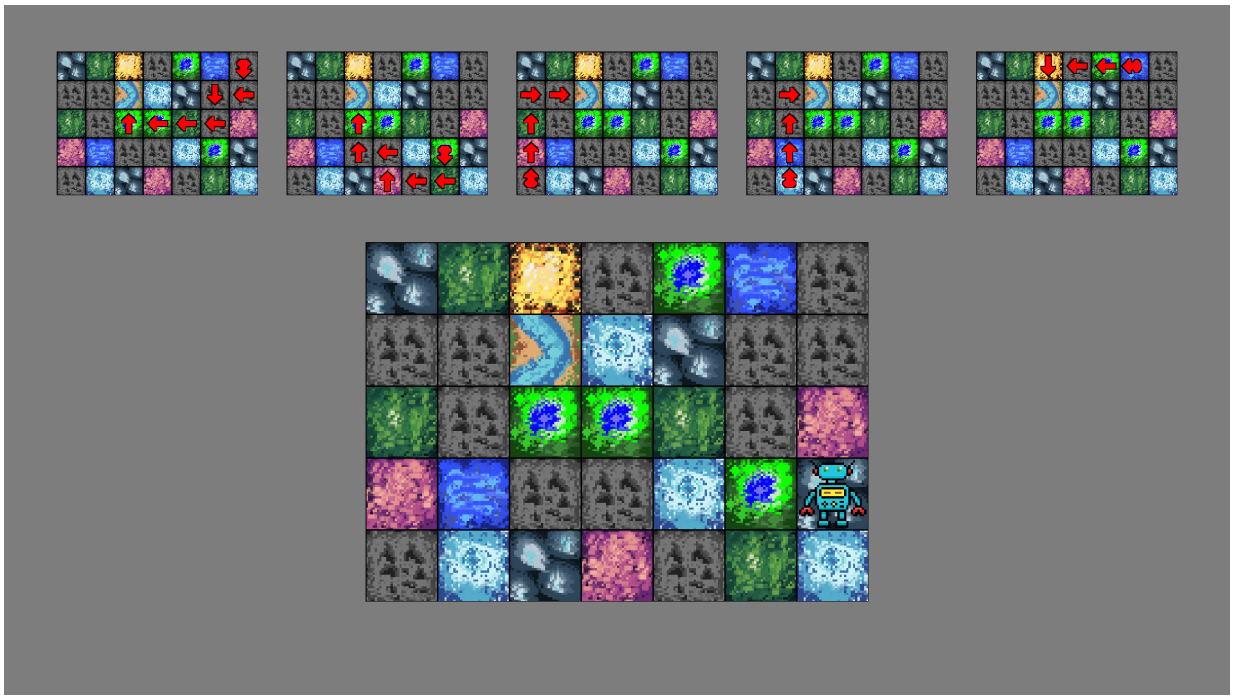
The study took place in a brightly lit room using a computer monitor that had a screen resolution of 2560 $\times$ 1440 pixels and dimensions of 595 $\times$ 335 mm. The monitor had a refresh rate of 60 Hz. The participants were seated and positioned in front of the monitor at an approximate distance of 110 cm. We recorded monocular eye movement data using the EyeLink 1000 Plus with a sampling rate of 1000 Hz. As Setup we used the we used the Remote Participant Setup provided by EyeLink. The Remote Setup is not using a headrest but a small target sticker placed on the participant's forehead, which makes it more comfortable for the participant. Additionally, we utilized a custom calibration target, recommended by Thaler et al. (2013), to optimize calibration accuracy and minimize eye movements during target fixation.
Note: Due to an implementation error we only recorded gaze data for the Transfer-Condition.

Software

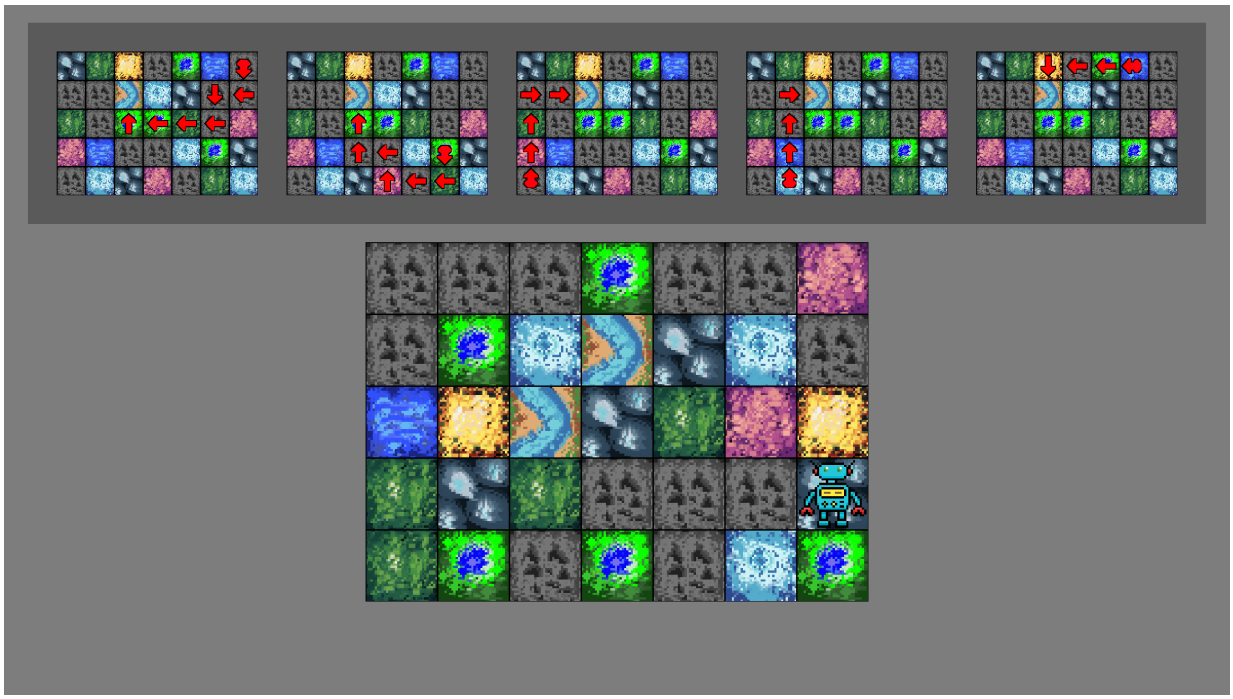
The experiment was implemented in Python (version 3.8.16) using PyGaze. The PyGaze toolbox is an open-source software package for Python (Dalmaijer et al., 2014). PyGaze can be used for visual and auditory stimulus presentation and provides an easy-to-use framework to communicate with eye trackers. For a complete eye-tracking experiment, at least two external packages are required: one for communication with the eye tracker and one for experiment processes. We used Pylink for communication and PyGame for the experiment process. For gaze- and event-data storage, we used the EDF file provided by EyeLink, which is then exported to an ASC file. For gaze detection, we used the software of the EyeLink tracker, which directly marks fixations, saccades, blinks, and other gaze events. For the analysis, we mainly used Pandas, Seaborn, and Matplotlib.

5.1.3. Experiment Procedure

Prior to the experiment, participants were asked to provide written consent and personal data were collected. Wearing contact lenses was allowed for participants. Also, eye dominance was measured before the experiment and the participant's dominant eye was used for recording. After participants read the instruction and chose a player name, we conducted a nine-point calibration and validation, until a validation score of average visual degree < 0.6 and max visual degree distance < 1, was achieved. The tracker was calibrated a first



(a) Base-Condition



(b) Transfer-Condition

Figure 5.1.: Image of an example trial from the experiment. Each trial consists of a Base-Condition and a Transfer-Condition, presented sequentially to the participant. Each condition screen is divided into a gridworld and a summary comprising five trajectories (smaller gridworlds on top). The arrows in the summary gridworlds represent optimal trajectories starting from specific start states, which are selected by the models. The task involves navigating the robot in the big gridworld to a goal state by maximizing the reward using the information provided in the summary.

time before the test trials and a second time after 16 main trials, to ensure better gaze data quality. For the same purpose, each trial started with a drift check, to reconstruct measurement errors and make sure that participants start each trial from the middle of the screen. After every condition participants got their performance displayed, as percent of the optimal reward. Every four trials, the player was presented with their current score in a high score table. This should intensify their intrinsic motivation for performing well in each trial. Studies show that intrinsic motivation is actually promoted by the presentation of the player's current performance as a high score (Halan et al., 2010). During the test trials participants could ask questions until the objective of the task was understood completely. Participants were allowed to do a break before the second calibration and were able to rest in between each trial for a short time.

5.2. Evaluation

Our experiment aimed to address two primary challenges. Firstly, we sought to determine the reconstruction model that participants employed when solving the task. Secondly, we aimed to investigate how the task structure influenced their preference for a particular model. To achieve this, we devised specific evaluation metrics that should serve as indicators of whether participants tended to lean towards utilizing Imitation Learning or Inverse Reinforcement Learning in their policy reconstruction. The carefully selected evaluation metrics were instrumental in developing our user study design. By carefully considering these metrics, we aimed to understand the factors influencing participants' model choices. Analyzing participants' responses and behaviors using these metrics further helped us gain insights into the factors guiding their model choices. In the following sections, we will delve into these metrics to provide a clear understanding of how participants chose their preferred reconstruction model.

Performance

We consider performance as our first evaluation metric with the aim of understanding the impact of different trials and task variations on human behavior and decision-making. To determine which reconstruction model participants adopted, we considered two performance features. We hypothesized that higher performance trials in the task could indicate the use of Inverse Reinforcement Learning by participants. IRL is likely to result in better performance due to its reliance on predicting optimal reward weights. Our computational experiment also supported this hypothesis, as IRL reconstruction consistently outperformed other combinations. Specifically, we anticipated that performance in the Transfer-Condition would serve as a critical indicator of model choice. In this condition, the summary does not perfectly match the world, and its purpose is to infer reward weights. Achieving higher performance in the Transfer-Condition would suggest that participants relied on IRL, as Imitation Learning would be less effective in this scenario.
Another informative indicator is the performance difference between the Base-Condition and the Transfer-Condition in a single trial. Using an IRL approach in the Base-Condition should lead to higher performance in the Transfer-Condition compared to using an IL approach. Moreover, if the performance in the Transfer-Condition is higher than in the Base-Condition, it could suggest that participants used imitation learning for the Base-Condition and inverse reinforcement learning for the Transfer-Condition. This is because only IRL should lead to higher performance in the Transfer-Condition since IL is not directly possible. These suggestions are also supported by our computational experiment results. Therefore, we propose that the performance difference between the two conditions is a crucial indicator for model choice. However, in general, we expect the performance in the Transfer-Condition to be lower.

Gaze Data

Lage et al. (2019) gained insight into the reconstruction model deployed by humans by directly asking them about their thought processes and how they used the summary afterward. In our study, we will take a different approach and measure human gaze behavior, which could provide valuable insights into their use of the summary. Studies have demonstrated that eye movements can effectively help understand human behavior and cognitive processes (Raptis et al., 2017; Reichle et al., 1998).
We hypothesize that the proportion of time humans spend looking at the summary could indicate a cognitive inference process, potentially reflecting an IRL approach. A higher proportion of time allocated to the summary may suggest that participants are inferring reward weights based on the given information. Conversely, shorter fixation times on the summary might indicate an IL behavior, where participants rely more on memorizing trajectories from the summary.
Additionally, we considered the number of switches between the summary and the gridworld as a potential indicator of the reconstruction model used by participants. A higher number of switches could suggest the use of IL, as participants might attempt to copy the trajectories presented in the summary to navigate the gridworld effectively. On the other hand, fewer switches could indicate an IRL approach, where participants rely more on the inferred reward weights from the summary rather than explicitly following the trajectories.

Summary Similarity

In addition to the previously mentioned measures, the evaluation of summary similarity, as defined in section 2.5, could serve as another indicator for the reconstruction method chosen by participants. It is important to note that this metric only applies to the Base-Condition, where the trajectories can be copied to solve the gridworld. We propose that if participants strictly adhere to the trajectories presented in the summary without any deviations, it would suggest the utilization of an IL approach. Imitation learners tend to closely mimic the provided trajectories, resulting in high similarity scores between their observed paths and the summary. However, it is essential to note that in some cases, the optimal path may include states from the summary, which can affect the accuracy of the measure. Therefore, we will solely focus on the differences between high and low manipulation of each variation and avoid relying on total similarity.

5.3. Trial Creation

As explained in chapter 3, we generated 50 sets of conditions, each consisting of two gridworlds with a shared reward weight. These condition sets were then utilized in the computational experiment to optimize an IL and IRL summary for each Base-Condition within each set. As a result, we compiled a comprehensive dataset of 800 trials, evenly divided into 400 trials featuring an IL summary and 400 employing an IRL summary. Additionally, we carefully selected a start state for each of these trials based on specific constraints, which will be explained in the following. To ensure a balanced and unbiased representation, the gridworlds were constructed randomly, incorporating only the different variation types. To prevent gridworld duplications and any potential recognition or bias from participants, we mirrored the IL trials. Following strict criteria, we proceeded to select 16 trials with IL summaries and 16 with IRL summaries for our subject experiment from the obtained set.

Base-Condition

The Base-Condition represents the initial gridworld presented in each trial (see figure 5.1a). In this condition, the participant is provided with a summary corresponding to the gridworld, which contains five smaller

gridworlds, each displaying a trajectory. The displayed summary for the Base-Condition is the one that achieved the highest reconstruction accuracy for this specific gridworld in our computational experiment. This approach is in line with the method used by Lage et al. (2019), ensuring that the human reconstruction can be as accurate as possible.
During our investigation and pilot studies, we discovered that the selection of the start state significantly influences our ability to determine which reconstruction model was used by humans. To strike a balance, we wanted the start state to be challenging enough, ensuring that participants would need to rely on the summary to find an optimal path. Additionally, we proposed that the path taken by participants in the Base-Condition could serve as an indicator of whether they used IL or IRL (cf. 5.2). Therefore, for each Base-Condition, we aimed to select start states based on the similarity of their optimal paths to the trajectories presented in the summary. Ideally, the optimal path from the selected start state should not include any of the states provided in the summary. With these ideas in mind, we established the following constraints for the start state selection:

- The start state cannot be a goal state.
- The start state must be at least two states away from any goal state.
- The start state cannot be one of the states included in the summary.
- The optimal path from the start state should not solely traverse regular-state dimension cells.
- The optimal path from the start state should have minimal similarity with the states provided in the summary.

However, due to the limited number of states in each gridworld (only 35), not all Base-Conditions could satisfy all of these constraints. Therefore, we applied the constraints in the specified order until only one suitable start state remained. Each outcome was assigned a ranking number, which later helped us choose the optimal trials for our experiments.

Transfer-Condition

The Transfer-Condition corresponds to the second gridworld presented in each trial (see figure 5.1b). It is constructed in a similar manner to the Base-Condition, with one distinction: the displayed summary does not correspond to the transfer world, but instead remains identical to the summary shown in the Base-Condition. Consequently, the reward weights and state dimensions remain unchanged, with only the arrangement of the gridworld differing.
In the Transfer-Condition, we also found that the selection of the start state is crucial for determining participants' model choices. Since the summary does not match the gridworld, we applied slightly different constraints compared to the Base-Condition. Our objective was to obtain a start state where the IL and IRL models would use different paths to reach the goal state. As with the Base-Condition, we assigned ranking numbers to each outcome to facilitate later selection. Considering these differences, we applied the following constraints for the start state selection in the Transfer-Condition:

- The start state is not a goal state.
- The start state is further than two cells away from a goal state.
- The optimal path from the start state should not solely traverse regular-state dimension cells.
- The path taken by the IL and IRL models differs for the start state.

Trial selection

After generating 50 IL and 50 IRL trials for each variation combination, we carefully selected the two trials that best met our criteria from each set of 50. To make this decision, we considered the combined ranking number of both conditions, covering both the IL and IRL trials, and selected the ones with the best ranking. As a result of this selection process, we ended up with a total of 32 trials: 16 with IRL summaries and the same 16 trials mirrored, featuring an IL summary.

5.4. Participants

A total of 22 participants took part in this study, including twelve females and ten males, aged between 18 and 39. All participants had normal vision or corrected-to-normal vision and provided written informed consent before participating in the experiment. They were explicitly informed about the collection of personal data, such as age, gender, click data, and eye movement data during the course of the experiment. Although the study did not offer monetary compensation, students from TU Darmstadt had the opportunity to accrue ”participation hours,” a requirement for Psychology and Cognitive Science students at the university. Initially, four participants served as pilot participants, helping to refine the experimental procedures. Ultimately, the final analysis incorporated data from 17 participants, as only complete datasets with adequate eye-tracking quality were considered.

6. Results - Subject Experiment

This chapter is dedicated to presenting the results of our subject experiment, focusing on three key aspects. Firstly, we will analyze the performance of the subjects, the time spent on the trials and some basic gaze data (section 6.1.1). Next, in section 6.2, we will investigate how variations in the task structure influenced performance, gaze and human behavior. Lastly, we will compare human performance and path behavior to the models, aiming to assess the validity of these models in predicting human extrapolation behavior (section 6.3). Through a thorough examination of these findings, we aim to gain valuable insights into the dynamics of our experiment and its implications for understanding how humans utilize policy summarizations.

6.1. Descriptive Analysis

To gain a comprehensive understanding of how individuals behave when using policy summarizations, it is crucial to analyze their overall performance and effort in our task. This approach allows us to discover important insights into the patterns and tendencies that emerge during the task, as well as the key factors that influence participants' decision-making processes. By exploring these aspects, we gain a better understanding of how people interact with policy summarizations in our study.

6.1.1. Performance

To assess the participants' performance, we utilized the proportion of optimal reward, as explained in section 2.5. This measure represents the ratio of achieved optimal rewards for each condition, and the trial performance is calculated as the mean of both conditions. Upon analyzing the mean performance of participants in each trial, we found that they achieved an average performance of 97% (mean=0.97, SD=0.06, min=0.43, max=1). Surprisingly, in nearly a third of the trials, subjects attained a perfect performance, with every participant accomplishing this level of performance multiple times.
In Figure 6.1, we present the mean performance of participants for each individual trial, color-coded to reflect the underlying variation combination and summary method. The first and third, as well as the second and fourth bars of one color tone, represent the same trial but with mirrored conditions and different summary methods. The different performances in the same trials suggest that the summaries may vary in their helpfulness. Generally, performance discrepancies between IL and IRL summaries are observed for specific variations, as the performance in IRL trials appears to be higher most of the time. A two samples t-test even shows trials with IRL summary to lead to significantly better performance overall ($t(1086) = -3.364, p = 0.0008$). However, there are exceptions, indicating that the summary type is not the sole factor influencing performance.
It is essential to highlight that the same combination of variations and summary types (indicated by bars with the exact same color) can result in different performances. This suggests that trial difficulty is not solely determined by the employed variations and summary method. But we also found that there was in general no compelling difference in performance between the Base- and Transfer-Conditions of on trial. After all, certain variations appear to lead to more challenging trials, as their overall performance seems to be lower

than others. Particularly, the second and fourth variation combination (HD_HF_LS; orange and HD_LF_LS; purple) appear to lead to lower performance. Nevertheless, it is worth noting that the confidence intervals for these trials are larger, indicating a larger variance in performance scores.
In general, the confidence intervals offer valuable insights, with some trials exhibiting significant performance variability, while others show less variation. Notably, trials with an IRL summary and a high number of distraction dimensions seem to have lower variability, suggesting they were relatively easier to solve.

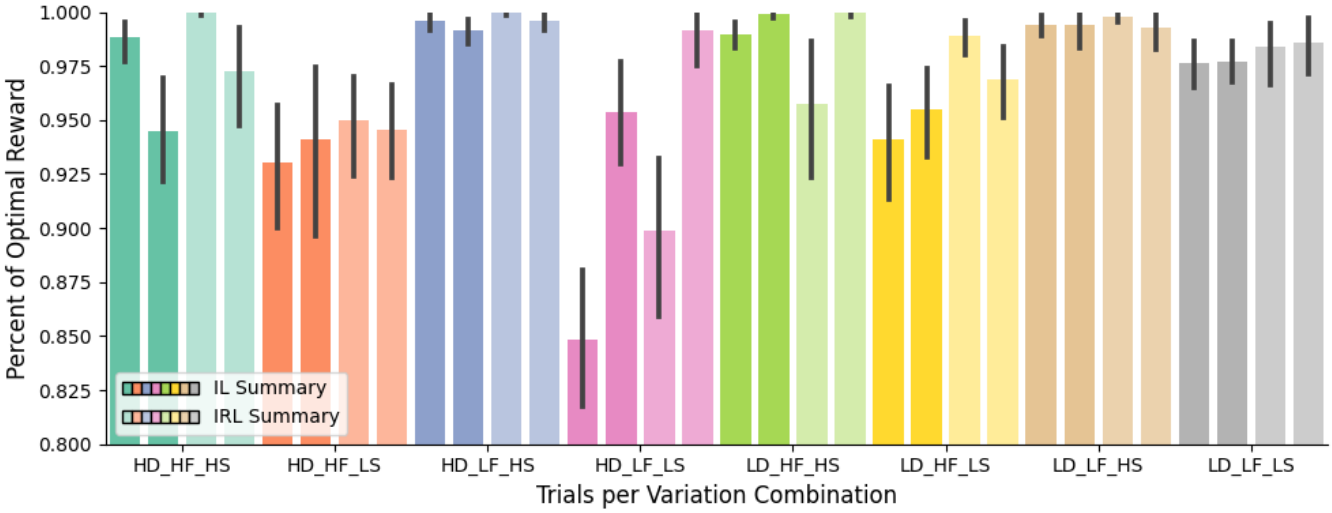


Figure 6.1.: Mean performance of all participants for each individual trial. Each bar represents a specific trial. The bars with the same color tone indicate identical variation combinations, while bars of the exact same color also share similar summary types (Lighter colors IRL; Brighter colors IL). The error bars displayed represent the 95% confidence interval. The x-axis labels represent different variation combinations, each denoted by abbreviations. ”HD” stands for high density, ”LD” for low density, ”HF” for a high number of state dimensions, and ”LF” for a low number of state dimensions. Similarly, ”HS” refers to a high number of distraction dimensions, and ”LS” to a low number of distraction dimensions. For example, ”HD_LF_LS” represents the variation combination where the gridworld has a high density, a low number of state dimensions, and a low number of distraction dimensions.

In order to comprehend the variation in performance shown in figure 6.1 and emphasize individual differences among participants, we refer to figure 6.2, which illustrates the mean performance per subject. It is worth noting that the mean performance of all participants is 97%, but the performance means per participant vary between 94.6% and 98.9% (mean=0.97, SD=0.012, min=0.946, max=0.989). This indicates noticeable differences among participants. The confidence intervals shown in the plot are particularly interesting. They reveal that participants' performance was not consistently the same throughout the trials, but rather exhibited a larger variance. Additionally, as the average performance increases, the confidence intervals become smaller. This suggests that a combination of skill and effort contributed to certain participants performing better than others.

6.1.2. Duration

Besides assessing performance, an interesting measure is the time spent on each trial and condition. This gives us valuable insights into the effort participants put into solving the trials and can also indicate how

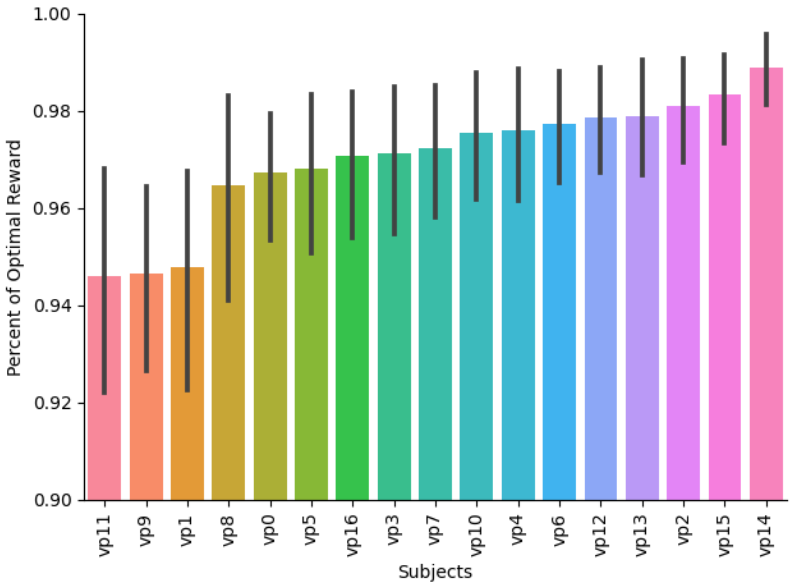


Figure 6.2.: Mean performance across all trials per subject, sorted in ascending order. The error bars represent the 95% confidence interval.

challenging a trial was. Additionally, it helps us understand whether participants genuinely tried to figure out the underlying rewards or simply followed the trajectories without deeper inference.
Figure 6.3 displays the average time required for each trial and condition, categorized into Base- and Transfer-Conditions. We observed that the mean time on trial varies strongly between each trial (mean=52.4, SD=17.3, min=23.9, max=100.2). Also on average, participants spent more time in the Base-Condition compared to the Transfer-Condition. The average time spent on the Base-Condition was approximately 29 seconds, which was about 0.3 times greater than the average time spent on the Transfer-Condition (23s). In general for some trials, the time spent on the Transfer-Condition seems to be substantially less than the time spent on the Base-Condition.
Interestingly, we found that the time spent on a trial did not strongly correlate with the performance achieved (compare figure 6.3 and 6.1). What is even more intriguing is that in some trials, we observed differences in time spent between IRL and IL summaries, even though they navigation task is exactly the same. Regardless, these differences did not follow any particular pattern, making them less predictable and suggesting that other factors might be at play.
On average, participants spent approx. 52 seconds per trial. The time spent on trials varied noticeably, with the participant who spent the least average time on the trial having 16 seconds, and the participant with the highest average time spent around 114 seconds. This substantial variation indicates a wide disparity in effort invested, with a standard deviation of about 27 seconds (mean=52.4, SD=27, min=16, max=114.4). An overview of the average time spent per participant can be seen in Figure 6.4. The plot provides a clear illustration of the notable variations in motivation and effort among the subjects. Individual levels of engagement can differ dramatically, resulting in considerable differences in the time spent on the trials.
Interestingly, when comparing the time spent on Base- and Transfer-Conditions per subject, the differences appear to be relatively similar, with the Transfer-Conditions being lower. This suggests that while participants' overall effort may vary significantly, their allocation of time between the two conditions tends to exhibit more consistent patterns. The plot also shows that more time does not mean a higher performance as the bars are arranged in ascending order of performance. This indicates that motivation and effort is not the only factor

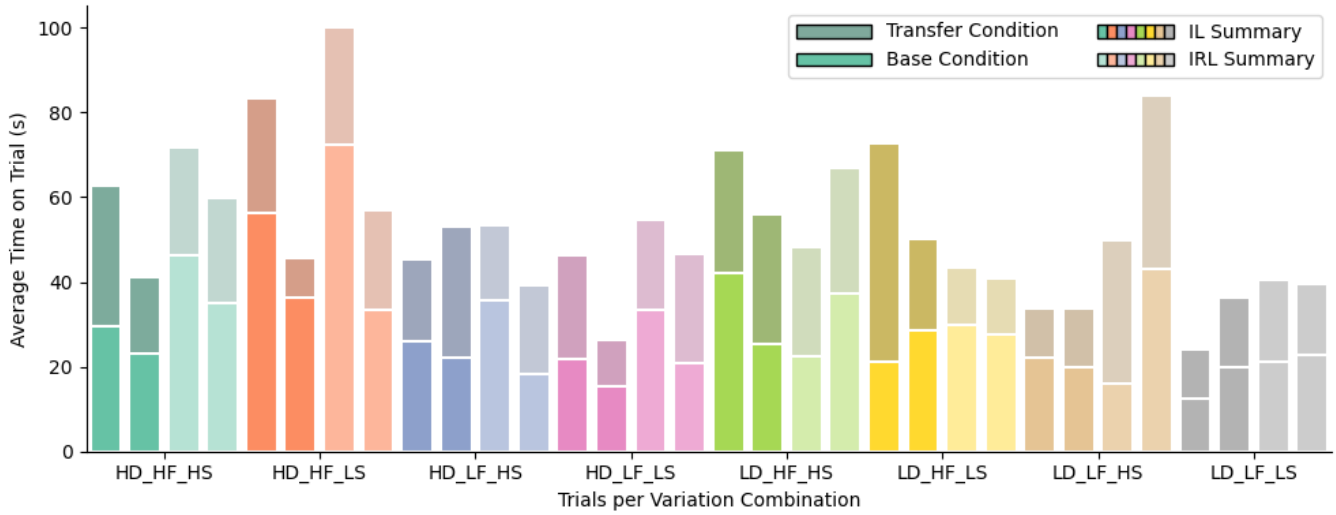


Figure 6.3.: Mean time per trial and condition. Bars of the same color represent trials with similar variations, and bars of the exact same color also have similar summary types (Lighter colors IL; Brighter colors IRL). Each bar shows the total time spent on the trial. The lower part of the bar indicates the time spent on the Base-Condition, while the upper part represents the average time spent on the Transfer-Condition.

influencing the participants' performance.

6.1.3. Gaze

Even though gaze data was only recorded for the Transfer-Condition due to implementation errors, we still managed to collect a substantial amount of data, totaling 38,838 fixations across all 17 subjects. Out of these, 21,564 fixations were on the gridworld, and 16,165 were on the summary. More detailed results about the fixations will be discussed in section 6.2.2. However, we will highlight one key point here, which are also depicted in Figure 6.5. The figure displays a heatmap of all the fixations recorded. Notably, the gridworld received a high density of fixations, indicating its importance to the participants.
Despite that, what is particularly intriguing about this plot is the fixation density on each summary trajectory. As we move from left to right, the fixation density decreases drastically. There are a couple of possible explanations for this observation. Nearly a third of the fixations on the summary was on the first trajectory. Humans typically read from left to right, which could explain the pattern seen in the fixation density. Alternatively, another interpretation is related to the trajectory positions on the summary. The trajectory on the left, being the one selected first by both the IL and IRL extraction model, provides the most information.

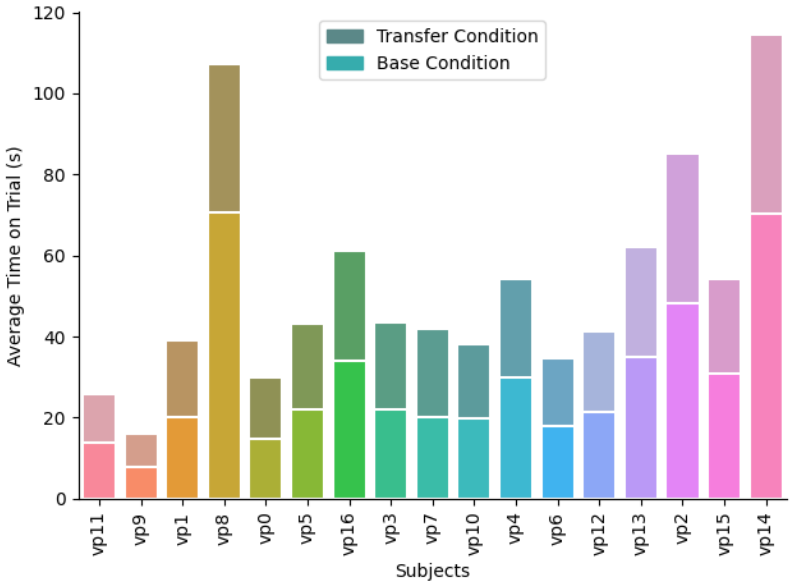


Figure 6.4.: Mean time spent on the trials per subject. Each bar indicates the average time spent on a trial per subject. The lower part of the bar indicates the time spent on the Base-Condition, while the upper part represents the average time spent on the Transfer-Condition. The bars are arranged in ascending order of performance, with the lowest performance on the left and the highest on the right

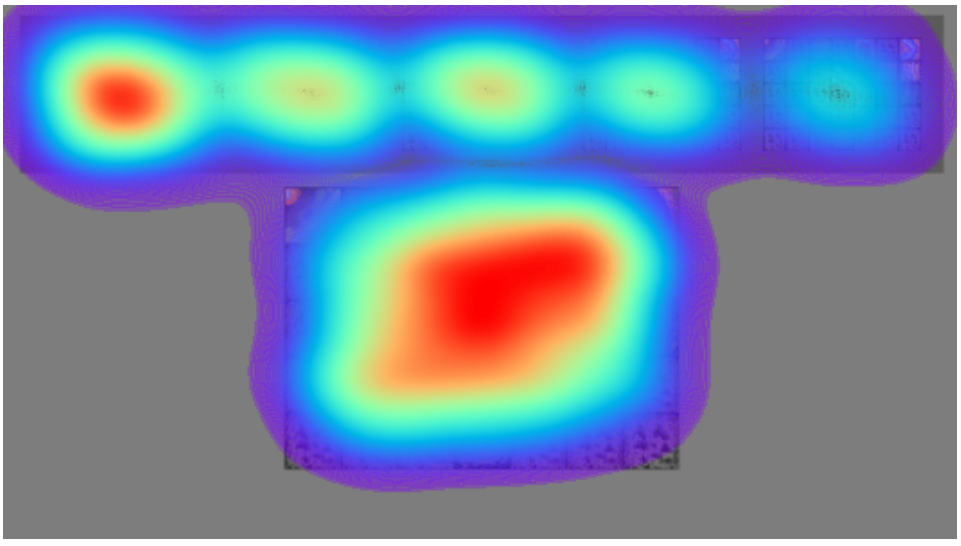


Figure 6.5.: Kernel Density Estimation (KDE Gaussian Kernel) Heatmap displaying the distribution of fixations of all participants. Warmer colors indicate a higher density of fixations.

6.2. Influence of Task Structure

One of our primary inquiries focused on investigating the potential impact of task structure variations on human behavior. Through the computational experiment, we observed that these variations did indeed affect the accuracy of the models, though to varying extents. In this section, we will present the results that could indicate these effects on humans. We will examine whether the variations influence the reconstruction model used by participants by analyzing not only their performance but also gaze data and the paths taken by individuals as they navigated through the gridworlds.

6.2.1. Subject Performance

In section 6.1.1, we observed varying performance ranges across the different trials; however, these deviations did not follow a consistent pattern. Nevertheless, we investigated whether the variations in task structure systematically influenced performance. Figure 6.6 presents a breakdown of the mean performance per variation. The figure clearly demonstrates performance differences among the various variations, particularly evident in the number of distraction dimensions. The disparities between high and low density are also readily apparent. Although the performance difference for the number of state dimensions variations appears to be relatively small, overall, these findings align with the results of the computational experiment. We further investigated whether a certain summary method led to better performance per variation but found that for all variations the IRL summary consistently resulted in better subject performance.

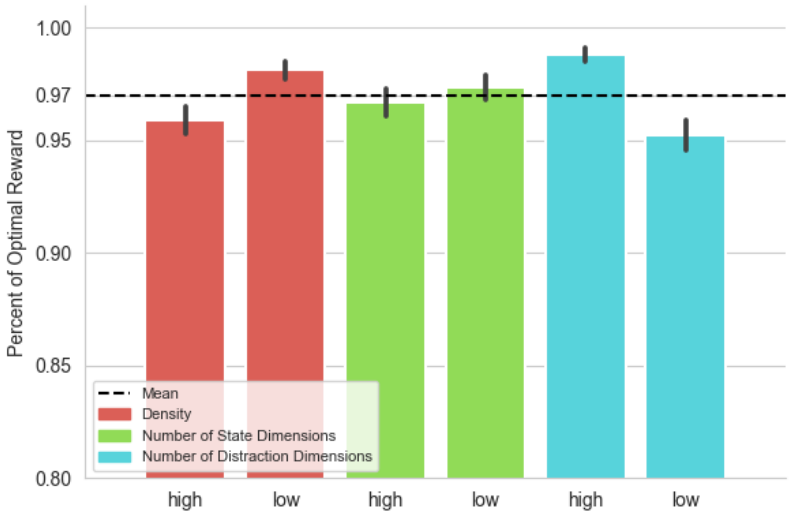


Figure 6.6.: Distribution of performance for each task structure variation across all subjects.

A more informative indicator for the deployed reconstruction model is the performance difference between the Base-Condition and the Transfer-Condition. Figure 6.7 presents the probability distribution of performance differences between the Base- and Transfer-Conditions. To be precise, it illustrates the distribution of performance differences for each trial played by each participant across all variations. It is noteworthy that for the Density variations, most participants exhibited a decrease in performance from the Base- to Transfer-Condition. Despite both high and low-density variations having a median difference close to 0, there were notably more negative differences for the low-density variation. Regarding the number of state dimensions,

a distinct pattern emerges. Both variations have a median performance difference close to 0, but the distribution of data points is significantly different. Surprisingly, a high number of state dimensions even led to an increase in performance between the two conditions. The variation with a low number of distraction dimensions appears to have the widest spread in differences, whereas the low state distraction variation demonstrates almost no spread at all, which corresponds well with the performance results. In conclusion, Figure 6.7 clearly demonstrates unmistakably differences among the various task structure variations, though the extent of these differences varies.

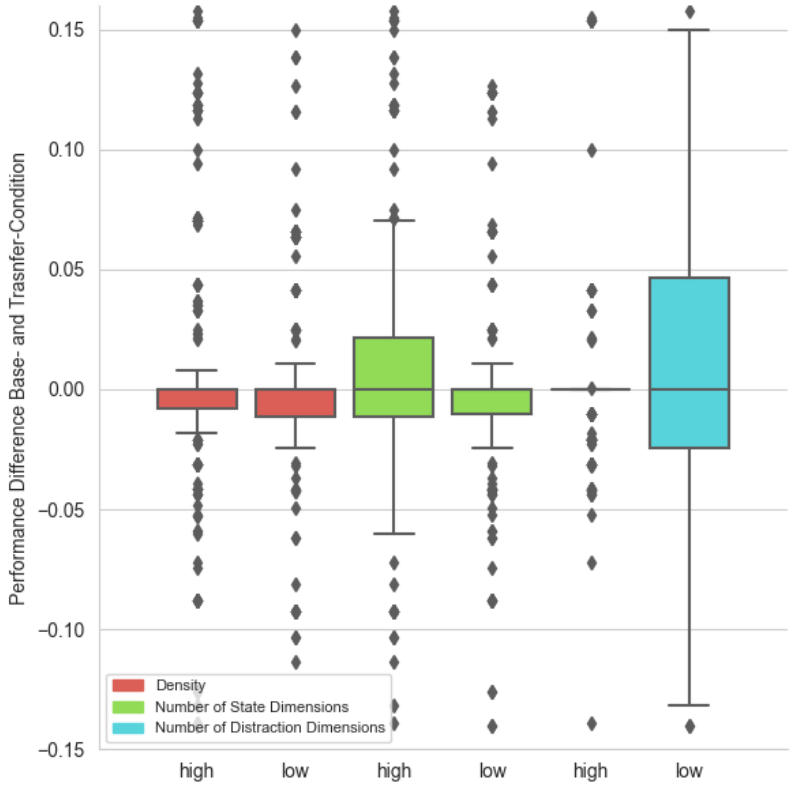


Figure 6.7.: Distribution of performance difference between Base- and Transfer-Condition of each trial per variation. Negative values indicate that the performance in the Transfer-Condition was worse than in the Base-Condition. Note that not all outliers are shown in the plot.

6.2.2. Gaze Behavior

Using the recorded gaze data, we investigate the frequency and duration of participants' gaze directed at the summary and their alternation between the summary and the gridworld. Due to data loss, we can only investigate the behavior in the Transfer-Condition and have therefore no baseline, but we can still see the impact of the variations in the Transfer-Condition. Figures 6.8 and 6.9 show examples of two participants' gaze behavior.

Particularly noteworthy are the differences in the length of the summary fixation periods. We believe that looking at the summary often but for short periods indicates a more IL-based approach. In contrast, looking at the summary for longer durations suggests that humans build a model and infer the reward weights. While the results shown in this plot are not entirely clear, we can distinguish these different behaviors for one

participant. For example, participant vp3 in Figure 6.8 seems to vary their strategy. In some trials, longer fixation times on the summary are used combined with fewer changes. In other trials, the gaze position is switched very often. As interesting as these insights may be, no clear pattern can be identified.



Figure 6.8.: Gaze behavior of participant vp3 for each trial over time. Each bar indicates one trial (or the Transfer-Condition of this trial). The darker areas on each bar represent the periods where the participant looked at the summary. The colored areas indicate the periods when they looked at the gridworld.

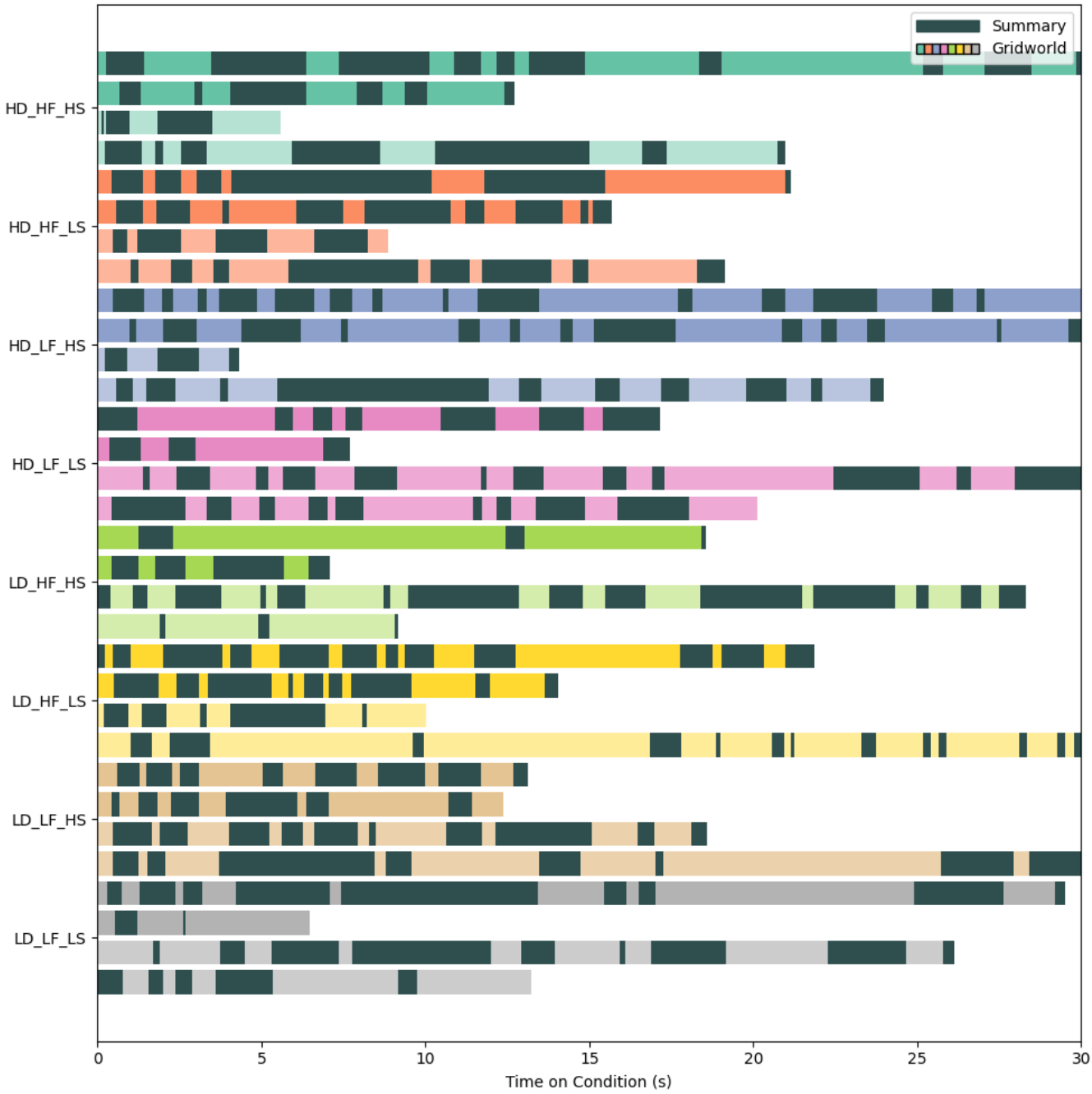
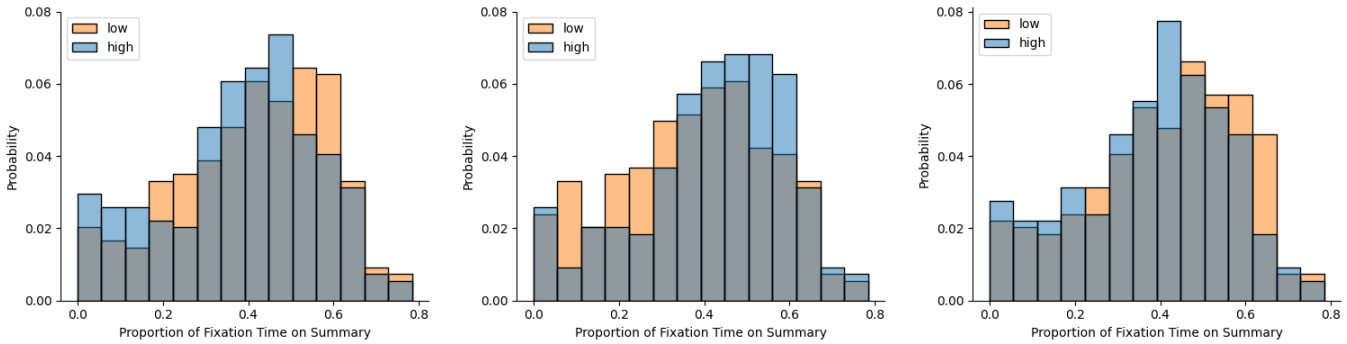


Figure 6.9.: Gaze behavior of participant vp7 for each trial over time. Each bar indicates one trial (or the Transfer-Condition of this trial). The darker areas on each bar represent the periods where the participant looked at the summary. The colored areas indicate the periods when they looked at the gridworld.

On the other hand, participant vp7 (Figure 6.9) appears to use a more IL-based approach. In total, the summary fixation periods are way shorter and way more frequent. This observation could indicate that there are model choice differences between participants. Nonetheless, it is crucial to acknowledge that even in the case of vp7, certain trials exhibited longer fixation periods, presenting a challenge when attempting to reach definitive conclusions. Equally intriguing is the examination of participant performance. It is noteworthy to mention that we observed two distinct approaches employed by these participants. Surprisingly, despite the contrasting strategies, their overall performance remains remarkably similar, as depicted in figure 6.1.
Considering Figures 6.8 and 6.9, there seem to be two indicators to assess the impact of task structure on the reconstruction model used by humans. These indicators are the fixation time on the summary and the number of saccades between the summary and the gridworld. The fixation time on the summary serves as an indicator of how long individuals spend inferring the rewards and building a model. On the other hand, the number of saccades between the summary and the gridworld could indicate whether participants used the summary as a sequential guide to navigating through the gridworld.

Fixation time on summary

Figure 6.10 displays the distribution of fixation time proportion on the summary for each variation. In general, we observe that all three distributions are somewhat normally distributed. Regardless, the most mesmerizing aspect of these plots lies in the colored segments highlighting the differences between the high and low variations. Though not entirely clear, we can discern some tendencies for each variation. For the density variation, lower variations seem to result in more fixation time on the summary compared to the higher variations, although not really evident. The same trend is apparent for the number of distraction dimensions. However, the most distinct pattern emerges in the number of state dimensions. Here, a lower number of state dimensions undoubtedly leads to a lower proportional fixation time compared to a higher number.

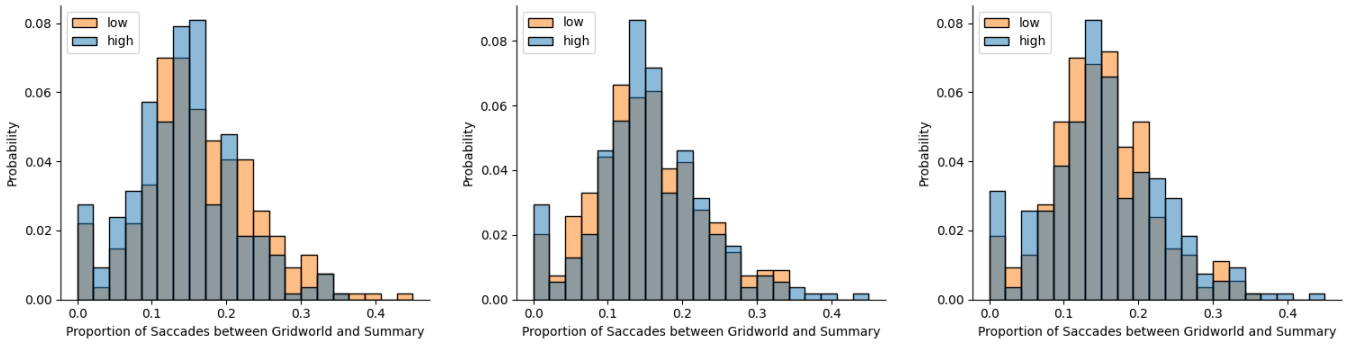


(a) Density **(b)** Number of State Dimensions **(c)** Number of Distractions Dimensions

Figure 6.10.: Distribution of fixation time proportion on the summary across participants and trials for different variations. The x-axis represents the proportion of fixation time spent on the summary, where a value of 0.2 indicates that 20% of the fixation time during the trial was directed at the summary. The grey area in each subplot indicates where the number of fixation proportions was the same, and the colored area above it illustrates the extent to which the variations (high or low) differed.

Saccades between summary and gridworld

Since we hypothesize that the number of gaze switches between the gridworld and the summary could be indicative of imitation learning behavior, Figure 6.11 displays the distribution of saccades between the gridworld and the summary for each variation type. Similar to the fixation time plots, these distributions also appear to be somewhat normally distributed. Yet, the differences between high and low variations seem to exhibit more randomness for the number of state dimensions and the number of distraction dimensions. In contrast, for the Density variation, the differences appear to be more pronounced, as the lower variation has more data points in the higher proportion area compared to the higher variation. These differences could suggest that certain variations impact participants' decision-making strategies and gaze behavior differently, which could potentially influence the extent to which they rely on imitation learning approaches.



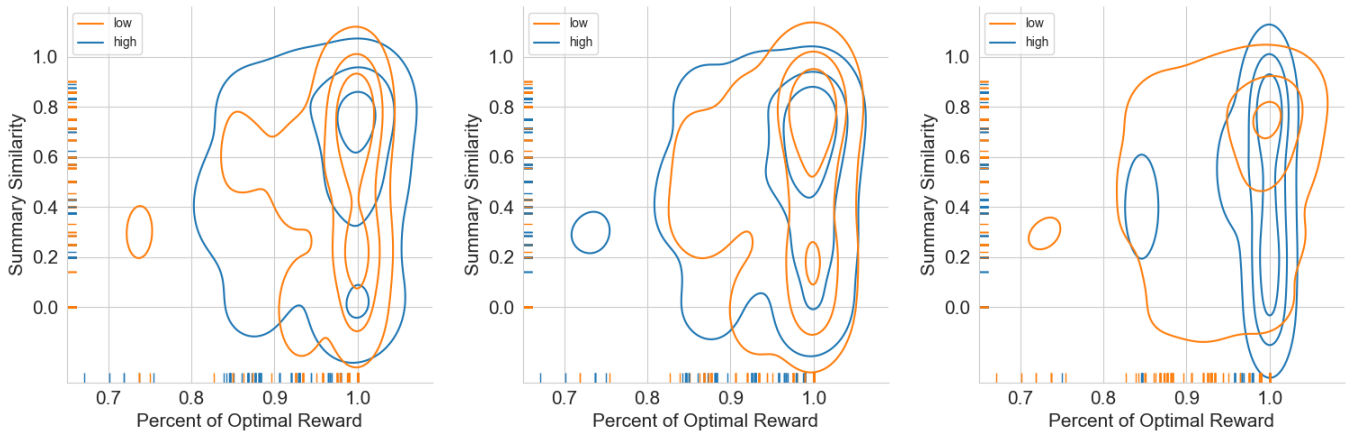
(a) Density **(b)** Number of State Dimensions **(c)** Number of Distractions Dimensions

Figure 6.11.: Distribution of saccade proportion between summary and girdworld across participants and trials for different variations. The x-axis represents the proportion of saccades made between summary and gridworld, where a value of 0.2 indicates that 20% of the saccades made in the trial were between gridworld and summary. The grey area in each subplot indicates where the proportion of the saccades was the same for both variations, and the colored area above it illustrates the extent to which they (high or low) differed.

6.2.3. Summary Similarity

Another measure we think could predict the reconstruction model deployed by humans is the summary similarity (cf. section 2.5) and section 5.2. As mentioned not in all trials the total summary similarity is a good indicator since often the optimal path even contains summary states. Therefore it is essential to only focus on the differences between the high and low variations.
Figure 6.12 presents a Kernel Density Estimation illustrating the distribution of summary similarity for each trial and participant across different variations. Notably, the most significant difference lies between a high and low number of distraction dimensions (figure 6.12c). In the low variation, the density appears to center around an approximate value of 75% summary similarity and 100% of optimal reward. In contrast, the high variation exhibits a more scattered distribution, encompassing data points spanning the entire range of the y-axis. This observation suggests that participants stuck more to the summary in the low variation compared to the high variation.
The opposite trend and a narrower range can be observed in Figure 6.12a. For the lower density variation, the data points appear to be scattered more evenly across all values of summary similarity, indicating a diverse range of approaches taken by participants. Conversely, in the high density variation, the majority

of data points exhibit high summary similarity, and they are centered around this value. There are fewer data points with very low similarity in this case, suggesting that participants in the high density variation tended to converge towards higher summary similarity values, leaving a smaller number of instances with significantly lower similarity. Regarding the number of state dimensions, there appear to be only slight disparities. Nevertheless, it is evident that the lower variation encompasses more paths with low summary similarity in contrast to the high variation (refer to figure 6.12b). This observation implies that participants in the lower variation were more likely to explore paths with lower summary similarity values compared to their counterparts in the high variation.



(a) Density **(b)** Number of State Dimensions **(c)** Number of Distraction Dimensions

Figure 6.12.: Kernel Density Estimation (Gaussian Kernel) over the summary similarity for the percent of optimal reward.

6.3. Can the Models Predict Human Behavior?

Since the IL and the IRL model were specifically chosen to simulate human extrapolation behavior, in this section we will present a comparison between these models and human participants in our task. Our aim is to examine the accuracy of these models in replicating or predicting human behavior in our task. By conducting this investigation, we hope to gain insights into the level of accuracy exhibited by these models and how closely they align with human performance and task solving methods.

6.3.1. Performance Comparison

In the computational experiment results, we primarily focused on evaluating the reconstruction accuracy of the models without considering their performance in the navigation task. To address this, we will now compare the performance of the models with the performance of the subjects. Figure 6.13 illustrates the disparities between the models' predicted performance and the actual performance for each subject, categorized by condition and model. On the whole, the predictions demonstrate a high level of accuracy. However, there is an exception with the IL model in the Transfer-Condition, which can be attributed to the loops discussed earlier (refer to section 3.2). This effect could have been further amplified by our trial selection, as we deliberately ensured that for the Transfer-Condition, the start state was chosen in a manner that led the

two models to follow distinct paths to the goal state. It is important to mention that the accuracy achieved may not solely result from precise predictions, but could also be influenced by the fact that both the models and human subjects exhibited a high level of reconstruction accuracy. Hence, their performance might not have been a mere prediction, but rather a proficient execution of the task in a manner similar to the human subjects. Nevertheless, the IL Transfer-Condition does indicate significant differences, leading to inferior predictions in this case.

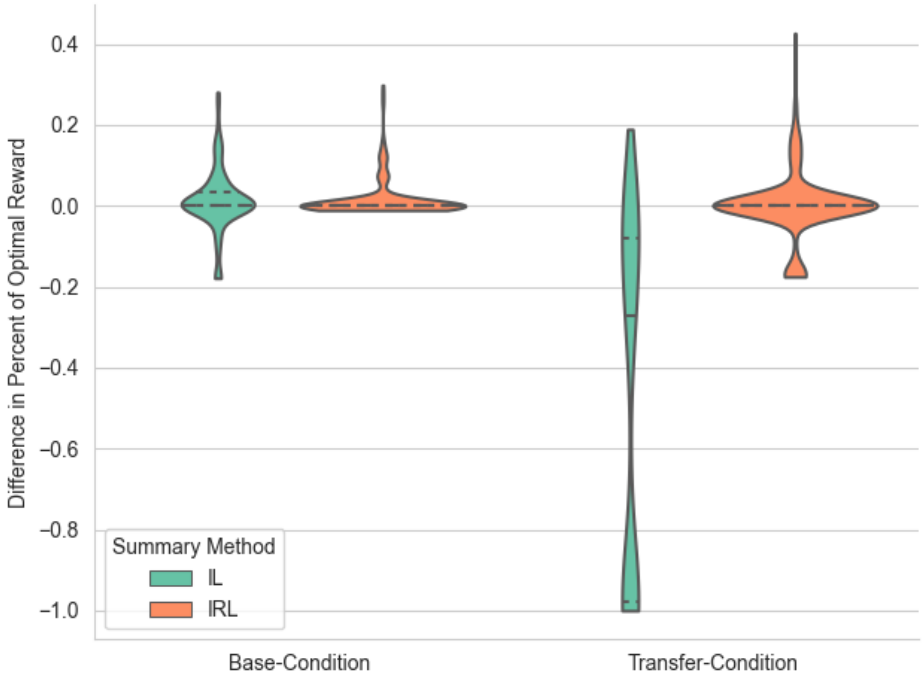


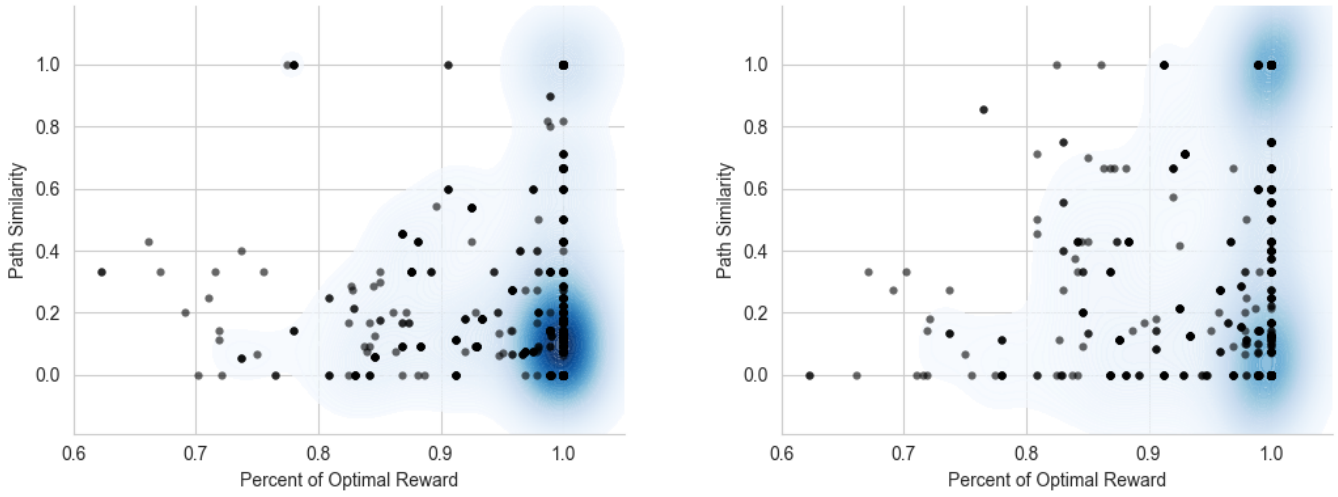
Figure 6.13.: Distribution of differences between model and human performance. Positive values indicate models outperformed humans, while negative values indicate worse performance compared to humans. For example, a negative value of -0.2 means that the performance was 0.2 lower, indicating that the model performed worse compared to the subject. The dotted lines inside each violin represent the median and quartiles.

6.3.2. Path Comparison

In addition to the performance comparison, we also explore how accurately each model predicts the paths followed by human participants. To assess this, we employ the path similarity measure described in section 2.5. Figure 6.14 showcases the path similarity between human paths and model predictions, organized by their respective performances. Subfigure 6.14a pertains to the IL model, while subfigure 6.14b represents the IRL model. Note that trials, where the model had a performance score of 0, were excluded from the analysis.

The plot reveals that the models struggle to accurately predict the paths taken by humans. While there are scattered instances where the predictions align with the actual paths, these occurrences appear to be more coincidental rather. The KDE demonstrates that the majority of paths are not correctly predicted by the IL model. On the other hand, the IRL model's predictions are somewhat divided, with approximately half of the paths being similar and the rest dissimilar. This suggests that the path similarity between humans and the IRL

model's predictions is generally higher, which could be attributed to the fact that humans use IRL or just that the IRL model is performing better in the task overall. It is worth noting that each condition may have multiple optimal paths, and the models tend to choose paths that seem to align best with the summary. Therefore, even if the participant found an optimal path the model path does not have to be similar. In conclusion, the path comparison indicates that the models' ability to accurately predict human paths is limited, and there is room for improvement in aligning model predictions with human behavior.



(a) IL **(b)** IRL

Figure 6.14.: Path similarity between subject paths and models. Trials where the models had a performance of 0 are excluded. The scatter points illustrate the path similarity for each condition and participant. The Kernel Density Estimation with a Gaussian Kernel shows the density of the scatter points as they are overlapped in some parts.

7. Discussion

The main focus of this research is to comprehensively investigate model-based policy summarization. Our study encompasses various aspects, including examining the accuracy of user models in reconstructing policies and exploring how different task structures influence this accuracy. Additionally, we aim to evaluate human performance and behavior in a policy summarization navigation task. Throughout our study, we explored the impact of task variation on human interaction and sought to determine the reconstruction models utilized by participants in different contexts. By addressing these research objectives, our aim is to contribute to a deeper understanding of policy summarization and its applications in diverse scenarios.
In this chapter, we will discuss the results from our experiments. We will first summarize the major findings from our computational experiment in section 7.1. Next, section 7.2 reflects on humans' extrapolation behavior when using policy summarizations to solve a task. Particularly, we will highlight the impact of task structure on human reconstruction methods in section 7.2.1. Lastly, we will discuss why the models used for policy summarization might not be the best choice and in what aspects they differ from human behavior (section 7.3).

7.1. Models of Human Extrapolation

Following Lage et al. (2019), we used an IRL-based and IL-based model to explore agent policy summarization for humans. Through a computational experiment, we gained interesting insights into their reconstruction accuracy and ability to model human extrapolation behavior. In the following section, we will discuss our main findings.

In our pursuit to validate the finding by Lage et al. (2019), which suggested that matching the extraction model to the reconstruction model yields high-quality reconstructions, we conducted similar computational experiments in a gridworld domain. However, our results reveal contrasting findings. Regardless of the extraction model used, we consistently observed that the IRL-based reconstruction model outperformed the IL-based reconstruction model. These discrepancies in outcomes are likely due to several computational differences:
One notable difference is the use of a distinct feature set for the IRL reconstruction. Initially, we employed their exact feature representation, resulting in comparatively lower reconstruction accuracy overall. Nonetheless, the performance gap between IRL and IL-based reconstruction models persisted, indicating that feature choice may not be the primary influencing factor.
Another aspect to consider is the summary size and its selection. While we used a fixed summary size of five trajectories for all summaries, Lage et al. (2019) employed varying sizes in their computational experiment. They showed that in the IL domain, more states in the summary generally improved reconstruction accuracy. However, for their gridworld domain, the summary size was chosen so that the best-performing IRL methods did not change in accuracy. Therefore this should not be the factor leading to these distinct outcomes. Furthermore, minor differences in hyperparameter selection are unlikely to have such a significant impact on the overall outcome.

The most significant divergence lies in the nature of the task itself. In our navigation task, the rewards were designed so an agent would find the best path to a goal state by maximizing the rewards. Conversely, the gridworld domain utilized by Lage et al. (2019) lacked goal or terminal states, contained 5 actions (including staying in place), and featured rewards not optimized for navigating to a certain state. As a result, trajectories in their scenario were less oriented towards reaching a specific goal state, resulting in a more random distribution of actions. This disparity in task characteristics could have influenced the Max-Ent algorithm, as matching the feature expectations (as described in section 2.3.1) was more straightforward in their gridworld, where no overarching trajectory planning was required. Consequently, we suggest that the Max-Ent algorithm was more suited to our task structure. This likely contributed to the substantial performance accuracy of an IRL model with an IL summary observed in our navigation task.
Despite these contrasting findings, our results align with human behavior, which generally favors IRL approaches for their superior knowledge transfer to diverse examples. In all cases, computational IRL tends to work better when summary sizes are of equal length. This observation is further evident in the Transfer-Condition, where the IRL-based reconstruction model consistently exhibited significantly better reconstruction quality than the IL-based model. This consistency indicates that successful knowledge transfer often necessitates a fundamental understanding and, in this task, an inference of the reward weights.
In conclusion, the likely divergence in task conception contributes to these different results. Nonetheless, this serves as an intriguing counterexample to their findings, highlighting that they cannot be universally generalized. Consequently, we assert that matching the extraction model and reconstruction model does not always increase reconstruction accuracy, disproving their previous conclusions.

Investigating the influence of task structure on human extrapolation, we also examined whether the task variations impact the models used to simulate human behavior. While we did not expect a strong influence due to computers not having limited short-term memory storage we still included these task variations in our computational experiment. Surprisingly, we even observed differences in the accuracy of certain variations. However, these differences were more pronounced for some model combinations than for others and somewhat the pattern varied between the combinations. Notably, in the Base-Condition the IL reconstruction model showed slightly better accuracy for a low density, a low number of state dimensions, and a high number of distraction dimensions, for both an IL and an IRL summary. On the other hand, the IRL reconstruction model demonstrated slightly better accuracy for a low density, a low number of state dimensions, and a low number of distraction dimensions. While the accuracy differences were not extreme, they appeared to be systematic enough to draw these conclusions.
Even though the Transfer-Condition had an overall worse reconstruction accuracy, we found nearly the same impacts of task structure. The only difference is that for the IL model a higher density seemed to lead to a higher accuracy than the lower variation which is in contrast to the Base-Condition. What is particularly surprising is the difference found between the models concerning the number of distraction dimensions. We intentionally chose this structure to investigate its impact on the summary and participants. To our surprise, we observed that participants found gridworlds with a high number of distraction dimensions much easier to solve compared to those with a low number. Treating distraction dimensions just like regular-state dimensions resulted in optimal accuracy for participants. However, for the models, this is not as straightforward since they only consider the feature vectors and not the whole environment. Consequently, the Max-Ent algorithm faces challenges in distinguishing between all state dimensions, as the rewards do not differ significantly. As a result, inferring the correct rewards from demonstrations becomes more challenging due to the reduced overall difference in the reward vector. On the other hand, for the GRF, the higher reconstruction accuracy for a higher number of distraction dimensions is also likely influenced by the feature vector. The GRF's feature vector includes the state dimension of the neighbor states, making it easier to label them as similar to the regular-state dimensions. As a result, the IL model can identify and therefore treat them as regular states,

leading to a higher reconstruction accuracy for the high number of distraction dimensions.
Through this investigation, we hoped to discern whether the task structure and therefore the corresponding summary influences the reconstruction approach employed by humans. Comparing the differences between human behavior and the models, we considered that the computational models have nearly unlimited memory for this task compared to humans, allowing them to handle the number of state dimensions and density without difficulty. The results that they are still influenced and to a different extend between IL and IRL reconstruction model suggest that the impacts on humans would be even greater.
We especially want to highlight the results for the IRL reconstruction, as the differences between IL and IRL summaries were most pronounced in this combination. The variations between each combination are much stronger when using an IL summary. This further confirms the preference of the inverse reinforcement learning reconstruction model for IRL summaries, indicating that they can provide a more concise summary of the global policy. In general, the inverse reinforcement learning model demonstrates a better grasp of the underlying reward weights and is less affected by these variations. On the other hand, the imitation learning model is more influenced by these variations because it only imitates and does not possess a comprehensive insight into the task. As a result, it struggles to handle the complexities introduced by these variations.
Overall, it is evident that our task variations do have a certain degree of influence on the models, although not to an extreme extent. This can be attributed to the computational power available to computers, which enables them to handle these variations effectively. Nonetheless, these findings indicate that different model combinations are influenced differently by various task variations, and further research is warranted to explore the intricacies of this relationship.

7.2. Human Extrapolation Behavior

To explore the impact of task structure on both humans and algorithms and to investigate how distinct task structures influence the selection of reconstruction strategies deployed by humans we conducted our subject experiment. Our main findings are discussed in the following section.

We examined the effectiveness of humans in using policy summarizations to solve the presented navigation task. The results of our descriptive analysis revealed that, on the whole, participants performed remarkably well, indicating their ability to effectively utilize the policy summarizations provided. Nonetheless, we did notice some inconsistencies in their performances, with certain individual trials posing more challenges and leading to decreased performance. We attribute these difficulties to two factors: First, some trials or conditions were inherently harder to solve, and second, certain summaries contained higher levels of uncertainty and did not fully represent all states or optimal behaviors.
Another intriguing observation was the significant variation in the time spent by participants on different trials, suggesting varying levels of difficulty. Surprisingly, we found no direct correlation between the mean time invested in a trial and the overall performance. Additionally, individual time effort varied enormously, but this variability did not seem to correlate with performance either. Thus, the time taken to complete a trial did not necessarily indicate how well a participant performed in solving the task. Despite the variations in difficulty levels, participants demonstrated an impressive ability to transfer policies from one condition to another, as no strong differences were observed between Base- and Transfer-Conditions.
The performance of participants appears to be influenced by their skills and abilities, as well by the quality and informativeness of the provided summaries. Based on our analysis, we conclude that the summaries were well chosen and indeed provided participants with valuable information. The substantial difference in performance between IRL and IL summaries further supports the notion that the former were more informative for participants. Additionally, some IL summaries exhibited uncertainties, meaning that they lacked

sufficient information about certain states or state dimensions. This observation aligns with the nature of these algorithms. IRL Machine Teaching, aims to capture all relevant information by selectively choosing trajectories with halfspace constraints that cover the policy. Consequently, the first trajectory in each IRL summary, being greedily selected, provided the most information.
On the other hand, IL Active Learning simply chooses the summary to maximize the reconstruction accuracy of the GRF algorithm, which might not always result in capturing all the relevant information. Therefore, we suggest that IRL summaries are generally more informative for humans due to their information-capturing ability. Additionally, we want to highlight that we did not find any evidence indicating that an IL method led to better results for certain variations, which raises questions about the necessity of matching extraction and reconstruction models.
Throughout our investigation, we observed variations in difficulty levels even within trials of the same variation combination, leading to differences in performance. However, in general, participants displayed an impressive ability to effectively utilize policy summaries, showcasing the potential of policy summarization for enhancing human-computer interaction. Moreover, we noticed individual differences among participants in their proficiency at using policy summarizations, indicating that personal factors and preferences may influence their effectiveness. Despite the overall high success of policy summarization, considering these individual nuances could be crucial for further improving the summaries.
In conclusion, our results highlight the promising prospects of policy summarization. To fully harness its benefits, it is crucial to remain attentive to individual differences and effort levels. Additionally, the quality of information included in a summary plays a vital role in enabling humans to achieve even better performance with policy summarization.

7.2.1. Linking Task Structure to Reconstruction Models

Understanding how task structures influence which reconstruction method humans deploy can help customize and therefore improve policy summarizations. Hence, the main goal of this study is to shed light on the connection between task structure and the utilization of different policy reconstruction approaches. In our experiment subjects played instances of a navigation task with different underlying structural variations. By analyzing performance, gaze patterns, and path data, our objective is to explore how these variations influence the selection of two reconstruction models. The results of the evaluation metrics were somewhat ambiguous and inconclusive, so no definitive or absolute conclusion could be drawn.

Density As expected, we found that participants tended to perform better in tasks with lower density, resulting in higher overall performance. Our results also indicate that in the Transfer-Condition, the low-density variation leads to worse performance compared to the Base-Condition, unlike the high-density variation. This observation could serve as an initial indication that in high-density tasks, participants are more likely to adopt an IRL approach, which allows for better knowledge transfer. The gaze data analysis presents inconsistent and inconclusive results for the density variation. Nevertheless, the saccade analysis suggests that in the low-density variation, participants utilize more of an IL approach, as there is a consistently higher proportion of saccades on the summary. Fixation time results neither confirm nor disprove this hypothesis. Moreover, the summary similarity analysis indicates that the summary is more frequently used in the high-density variation, suggesting an IL model preference. Given the inconsistencies in our results, we cannot definitively conclude whether density significantly impacts model selection.

Number of State Dimensions For the number of state dimensions, a notable performance increase was observed from the Base-Condition to the Transfer-Condition, particularly for the high variation. This performance improvement can be attributed to two factors. Firstly, it is likely that participants learned from

their mistakes in the Base-Condition, resulting in enhanced performance in the Transfer-Condition. Secondly, the possibility that they just utilized the summary more effectively in the Transfer-Condition, which is less probable. Hence, based on these observations, no definitive statement can be made regarding any model preference. In this variation, the proportion of fixation time does indicate that, in general, more often more time was spent on the summary in the higher variation. Conversely, in the low variation, a clearly lower proportion of time was allocated to the summary. These findings suggest a tendency towards an IRL approach for the higher variation. However, the saccade analysis, in contrast, did not reveal any distinct tendencies. The summary similarity analysis did not reveal any significant differences either, leading us again to the conclusion that there is different behavior, but no reconstruction model can be assigned definitively.

Number of Distraction Dimensions We observed that the distraction dimensions had the most significant influence on human performance, primarily because a high number of distraction dimensions appeared to be too easy for most participants. In this variation, the performance was consistently close to the maximum, as no significant mistakes could be made by merely taking the shortest path. These findings indicate that the summarization algorithms effectively highlight the relevant states and avoid confusing participants with distractions. Moreover, it demonstrates that humans are quite robust when it comes to dealing with distractions and can even use them to their advantage. Nevertheless, in terms of linking a reconstruction model to these variations, the performance and performance differences do not provide clear hints. The gaze analysis also did not yield clear results either, as both the proportional fixation time and saccade proportion showed ambiguous patterns. On the other hand, the summary similarity results displayed more apparent trends, indicating an overall higher summary similarity for the lower variation. This suggests that in trials with lower distraction dimensions, more states of the summary were utilized, pointing towards a tendency of imitation learning. Nonetheless, the observed differences and overall results are still not definitive enough to provide absolute indications.

Unfortunately, we were not able to clearly link the reconstruction models to task structure despite seeing some tendencies. Nonetheless, we discovered that these variations do impact our computational user models and also influence human behavior. These outcomes could suggest several possibilities: It is possible that there are no distinct differences between the reconstruction models, which is unlikely and contradicts the findings of (Lage et al., 2019). Another scenario could be that our task structure does not have a significant influence on the choice of reconstruction models. Alternatively, our analysis may not be sensitive enough to detect such effects. It is important to consider that there might be individual differences among participants, leading to the possibility that different individuals consistently use different reconstruction models or vary their approach across different trials. This variability makes it challenging, and even impossible, to establish a general link between the task structure and a specific reconstruction model. This observation however raises the idea of personalized policy summaries tailored to individual participants' preferences and behaviors.

Furthermore, it was observed that humans generally spend more time in the Base-Condition compared to the Transfer-Condition, but this difference in time allocation did not result in a significant performance gap. This finding could suggest that humans might be employing an IRL approach and building a model during the Base-Condition all the time, which they can effectively transfer to the Transfer-Condition. Using a straight IL approach would not yield the same level of performance, as indicated by the computational analysis. Taking this into consideration, it appears challenging to clearly distinguish between different reconstruction model choices, which could explain our results and lead to the notion that humans may not be strictly utilizing IRL or IL, but a combination.

In conclusion, we were unable to establish a direct link between the reconstruction models and task structure. Nevertheless, this study revealed several ideas and suggestions that highlight the complexity but also the significance of the problem.

7.3. Differences Between Model Predictions and Human Behavior

Having conducted a computational and subject experiment we compared the models' predictions and human behavior (cf. 3.3). Although the models demonstrated relatively close approximations to human performance, they fell short in accurately predicting the specific paths chosen by humans.
The observed similarity in overall performance between the models and humans (cf. section 6.3.1) can be attributed to both groups achieving high levels of performance in the task. Therefore, we want to point out that the models may not be actively predicting human performance but rather reflecting the general high competence demonstrated in the task. Notably, in the Transfer-Condition, the IL model's performance significantly diverged from human behavior due to issues in reconstructing the underlying policy. This discrepancy highlights the failure of the IL model in precisely capturing human behavior.
Moreover, comparing the paths chosen by the models with those of humans revealed that the selected models do not accurately simulate human conduct. Any similarities identified in this analysis can largely be attributed to the fact that both humans and models tend to select optimal paths. It is essential to recognize that the chosen models are designed to represent mathematical principles related to imitation and inverse reinforcement learning, rather than being faithful models of human behavior. Therefore, they may not entirely capture the complexities of how humans make decisions. In reality, human decision-making often combines aspects of both imitation learning and inverse reinforcement learning, influenced by various heuristics and factors that do not strictly align with these mathematical models.
In conclusion, we found that both IRL and IL models have limitations when it comes to accurately predicting human behavior for several reasons. Human decision-making is not confined to a singular approach but rather involves a combination of different learning strategies. Additionally, human behavior is not always as rational and systematic as these algorithms, making it challenging for the models to fully grasp human behavior. Although the current models may not precisely predict human behavior, they still serve as valuable representations of distinct learning approaches, aiding our understanding of these learning mechanisms. However, it is crucial to acknowledge that these models cannot be seen as direct simulations of human-like behavior due to the inherent complexity and nuanced nature of human reasoning.

8. Directions for Future Research

Throughout the analysis, we identified certain limitations in our implementation and gained insight into the potential avenues for expanding our research. This section will explain possible ways to enhance the experimental setup, explore model improvement ideas, and discuss opportunities to further extend the research.

8.1. Experiment Improvements

Since our experiment is entirely self-developed, we have numerous ideas on how to improve it. Overall, we believe the experiment provides a solid setup for exploring our objectives, but there is room for further enhancements to gain clearer and more profound results.

Environment In general, we recommend selecting a more challenging environment to create a larger performance difference between participants, possibly prompting them to use different reconstruction methods. One way to achieve this is by increasing the size of our gridworld, making it harder to fully understand the summary. Alternatively, we could explore different implementations of our variations to increase the difficulty. Other options include using a non-deterministic environment, adding actions, or incorporating additional features that impact the agent's behavior. For example, adding walls or other terminal states could be interesting. We have even considered using pick-up items to further increase the complexity of both the summaries and the tasks. While it is generally a good idea to use a different task, we believe sticking to an MDP structure and a navigation task is reasonable since it closely resembles real-world problems.

Feedback and highscores Based on our findings, we noticed that humans learned from the Base-Condition and applied that knowledge in the Transfer-Condition. However, this outcome did not align with our initial idea of transferring. To address this issue, we recommend not providing feedback after the Base-Condition, as it may eliminate these disturbances and encourage a more accurate assessment of transferability between the two conditions. Moreover, we noticed that the highscore list greatly motivated certain participants, leading to extended trial durations. Conversely, others were less affected and didn't show the same level of motivation. To ensure a consistent comparison, it would be beneficial to find a way to equally motivate all participants during the study. Implementing a time limit could also be useful to prevent participants from taking an excessive amount of time on individual trials.

Task structure variations Making adjustments to our chosen variations would be beneficial to further improve the experiment. Specifically, we recommend modifying the number of the distraction dimensions variation. While including distraction dimensions is interesting, we suggest significantly reducing the probability of a dimension being a distraction dimension in the high variation condition to prevent trials from becoming too easy. Additionally, exploring further task structure variations could be intriguing to see how they impact the reconstruction models used. Further variations could be the number of goal states, variance in rewards,

time-dependent rewards, bottleneck states, and stochastic transitions. These variations could also provide valuable insights into the complexity of the tasks and their influence on human decision-making.

Summaries Firstly, we can change or vary the summary sizes to investigate how they influence reconstruction performance. Additionally, we could limit the summaries to a specific number of states. In our task, summaries could also include trajectories that do not reach the goal state, making it more interesting to see which summaries the models select. Furthermore, it would be intriguing to change the trajectory order during the experiment to determine if subjects prefer trajectories with more information. By exploring these changes and improvements, we could gain deeper understanding on the impact of different summary characteristics on human behavior and model performance.

Informativeness and uncertainty of summaries One aspect that we didn't consider in our research is the informativeness and uncertainty included in the summaries. As mentioned earlier, some summaries presented uncertainties by not including information about certain states. Measuring the informativeness and uncertainty for each summary could be highly interesting in understanding their helpfulness to humans. A useful measure for this is the BEC area, which is also utilized by the IRL Machine Teaching algorithm (see section 2.4.1). In a similar study that employed this measure, Lee et al. (2021) found that the information class of summaries had a significant impact on how humans perceived the informativeness. Interestingly, the perceived informativeness decreased from low to maximum information class. This suggests that the inherent information content of a demonstration, as measured by its BEC area, may not always align with the information effectively transferred to human learners. Moreover, they observed significant effects between information class and measures of mental effort or puzzlement. Incorporating these findings and further investigating and validating them could be a valuable improvement and idea for future studies. Understanding how informativeness and uncertainty in summaries influence human learning and decision-making can provide essential revelations into designing more effective policy summarization processes.

Analysis With respect to our data, further examination could be conducted, with a particular focus on the differences between individual participants. Additionally, more extensive gaze data analysis could provide valuable insights. For example, comparing which states were observed before and after examining a specific trajectory could shed light on participants' behavior in the task. Moreover, recording gaze data during the Base-Condition could enable the exploration of many more interesting analysis methods. The additional data would offer a deeper understanding of participants' decision-making processes and their interaction with the policy summarizations during the learning phase. Such data could greatly enhance our comprehension of human behavior and decision strategies in the navigation task, leading to more comprehensive and insightful results. But apart from our particular experiment, there is also room for improvement as our results indicate the challenge of fully understanding the reconstruction model humans use. While we think eyetracking is generally a good approach, in our case, it did not lead to clear results, probably due to various factors. Lage et al. (2019) obtained information about the used reconstruction model by directly asking the participants. While this is a promising approach, it should not be the sole indicator, as humans often do more things than they consciously notice. Therefore, a study where brain activity is measured (for example, using an EEG) could be very interesting. Exploring the brain's neural signals during the decision-making process might provide a deeper understanding of how humans comprehend and use policy summarizations, offering a more comprehensive understanding of their strategies.

8.2. User Models

The models used in our study have demonstrated the ability to provide effective summaries for agent behavior in our specific task. Notably, we observed that the IRL-based extraction outperformed the IL-based extraction in generating better summaries. Therefore, we recommend that future research reconsiders the use of an IL-based model, as we found that simply matching the models may not necessarily result in the highest reconstruction accuracy. Instead, we propose that the key factor lies in the informativeness of the summaries, making the IRL approach superior to the IL approach in this context. But still, our results contradict the findings of Lage et al. (2019), prompting us to suggest further exploration of the importance of matching extraction and reconstruction models. Conducting a new subject study to investigate this aspect, rather than relying solely on a computational experiment, could be beneficial. Additionally, given the absence of evidence supporting the superiority of an IL extraction for any of our variations, it becomes imperative to reexamine and confirm these findings in the first place.

Moreover, our findings indicate that both reconstruction models do not fully capture human behavior but rather represent the underlying approach. This insight prompts us to explore new user models that can better emulate human behavior. One potential avenue is employing neural networks to bridge the gap between IRL and IL, which reflects the reality of human learning. Adopting such user models provides an opportunity to focus on individual differences among participants and enable learning while humans interact with them to generate summaries. By incorporating neural networks into the user models, we could create more dynamic and adaptive systems that learn from human interactions, leading to more accurate and personalized summaries. An alternative approach would be to incorporate computational costs into the algorithms. For instance, the algorithms could not only consider the best reconstruction but also the one that strikes the best balance between accuracy and computational effort. This would lead to a more accurate modeling of human cognition. Considering these ideas could open up new possibilities for understanding human learning and decision-making processes and contributes to the development of more effective AI-assisted systems.

9. Conclusion

In this work, we conducted a computational and a subject experiment to investigate model-based policy summarization. Our main objectives were to validate the findings of Lage et al. (2019), which emphasized the effectiveness of matching extraction and reconstruction models for accurate reconstructions. Additionally, we aimed to investigate the impact of task structure on both human and model behavior, as well as how different task structures influenced the reconstruction strategies employed by humans. Lastly, we explored the models' predictive ability concerning human conduct, with a specific focus on the transfer of learned information.

To accomplish these objectives, we devised a navigation task that required participants to determine an optimal path using policy summarizations and transfer summary information between two conditions. Additionally, we incorporated three task structure variations, expecting them to influence the reconstruction model deployed by human participants.

In the computational experiment, we found that matching the extraction and reconstruction model did not increase reconstruction accuracy for our task, which contradicts the findings of Lage et al. (2019). Additionally, we observed a clear impact of task structure on reconstruction accuracy and a strong indication that an inverse reinforcement learning model performs exceptionally better in transferring information.

In the subject study, we conducted performance, path, and eye movement analyses, revealing that humans efficiently utilize policy summarizations to solve the task and, in numerous cases, even outperform the models. Discovering the specific reconstruction model employed by human participants proved to be challenging. Nevertheless, we found evidence for a connection between model choice and task structure.

We propose that the selected models might not completely capture the nuances of human behavior, and they might not be adaptable to individual differences among subjects. Hence, future research should prioritize refining these models or even exploring entirely new ones. Incorporating various aspects of human behavior and individual preferences into the models, rather than strictly distinguishing between imitation learning and inverse reinforcement learning, could lead to more accurate results. Additionally, conducting further investigation is essential to comprehend the significance of matching extraction and reconstruction models for human reconstruction, given the presence of varying results.

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A. Appendix: Computational Experiment

Model Performance

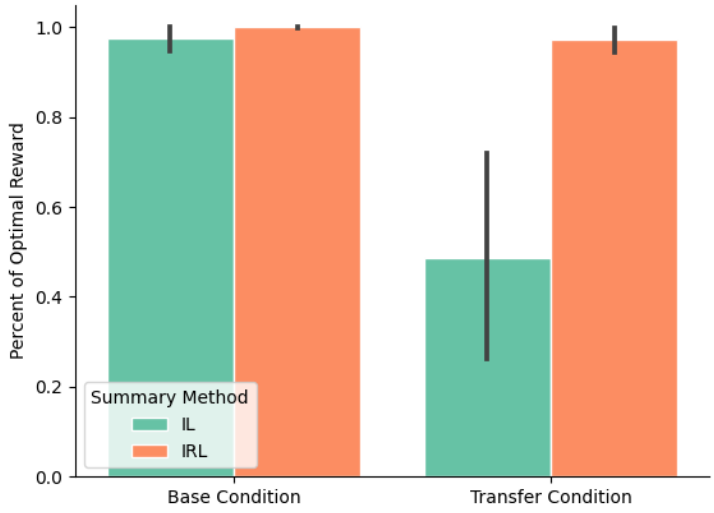
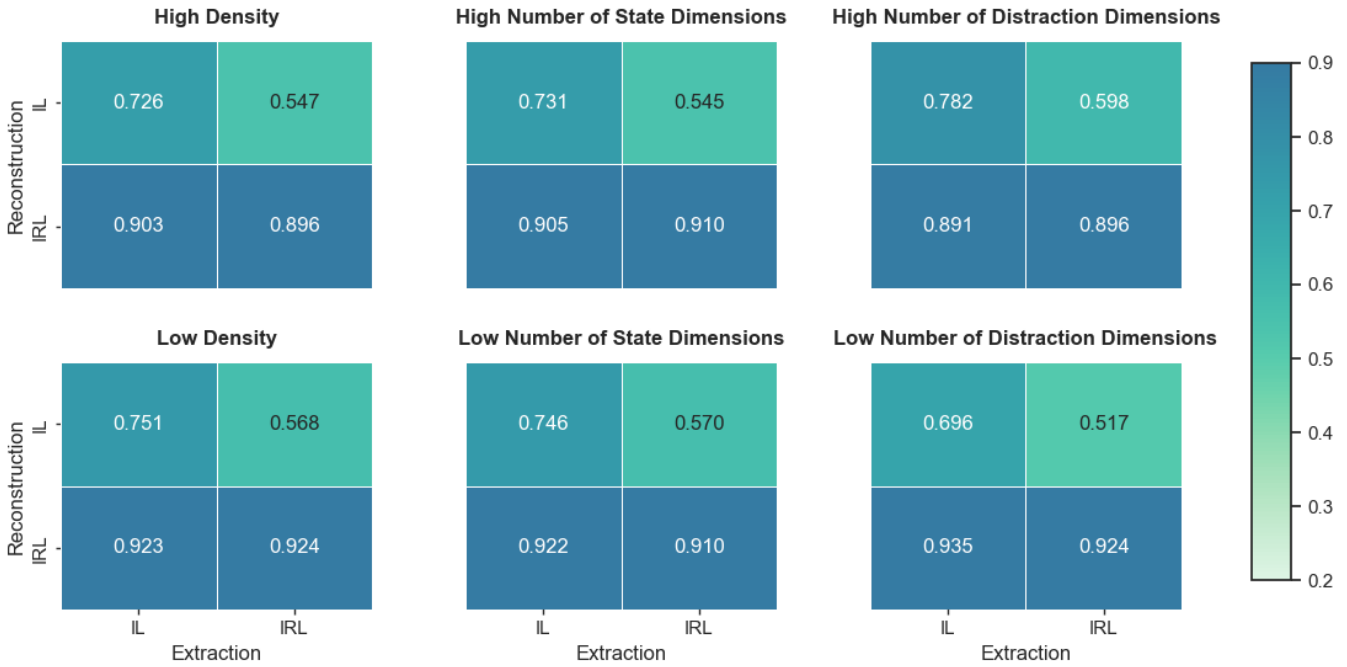
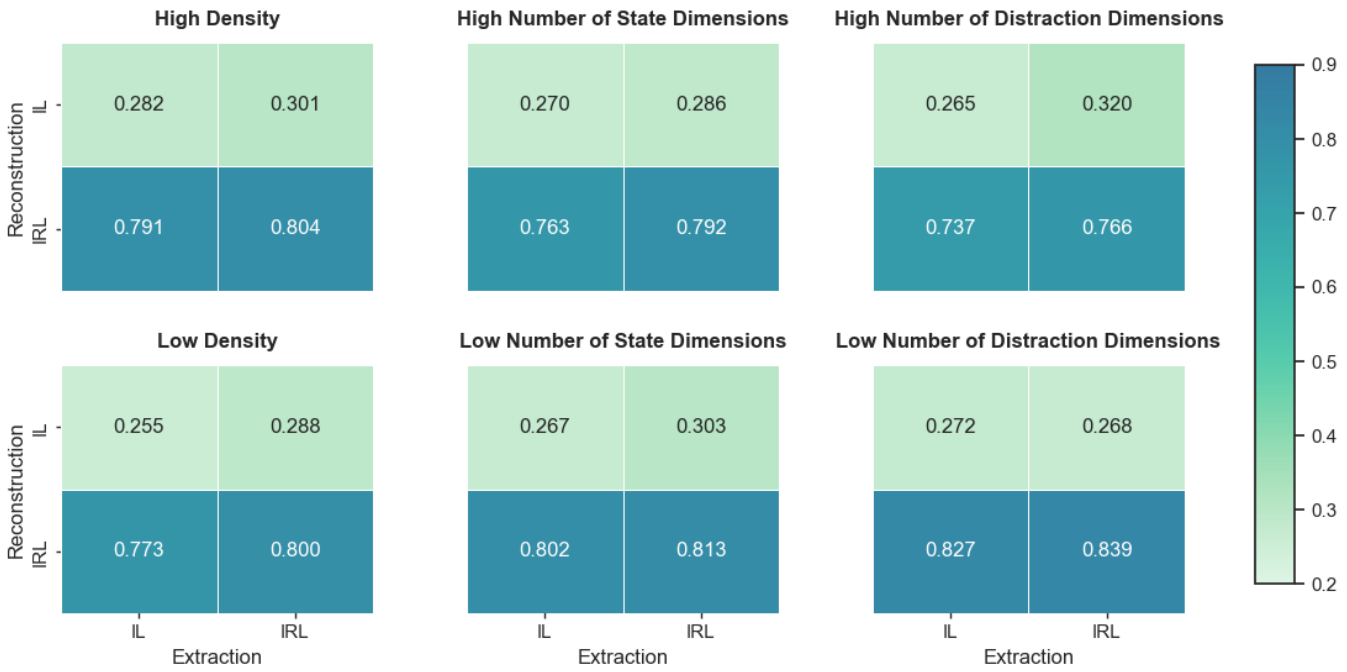


Figure A.1.: Mean model performance for Base- and Transfer-Condition per summary method. The error bars represent the 95% confidence interval.

Reconstruction Accuracy

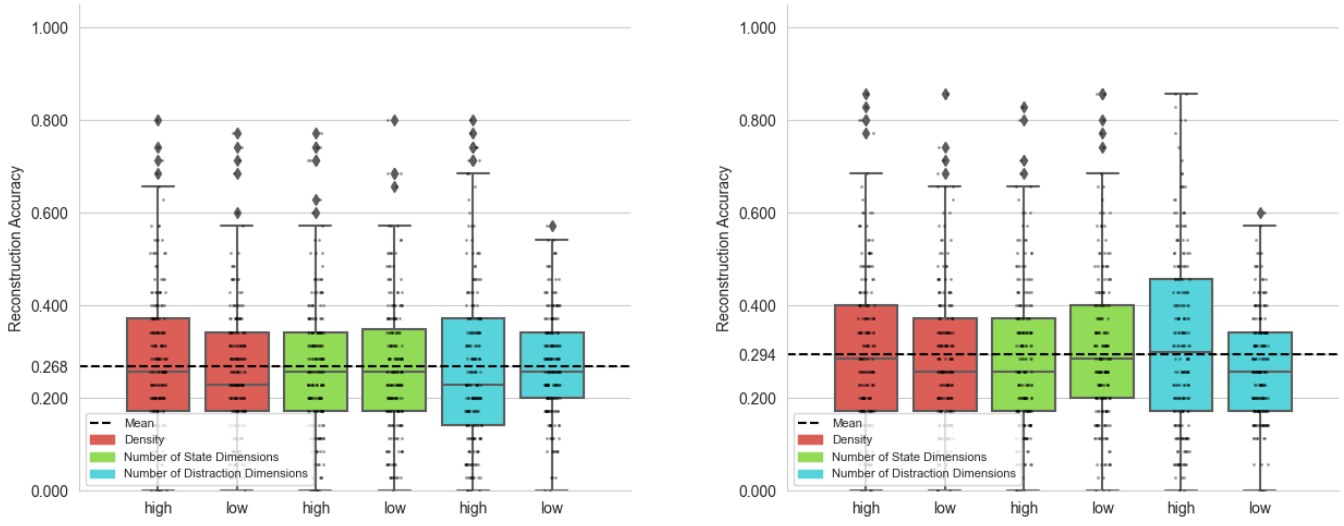


(a) Base-Condition



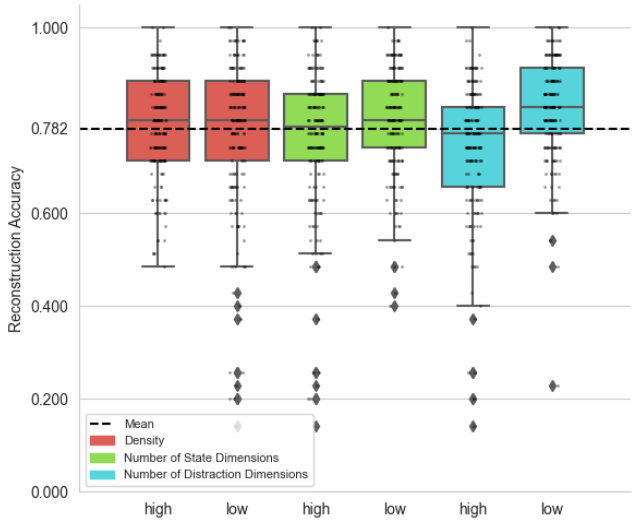
(b) Transfer-Condition

Figure A.2.: Mean reconstruction accuracy for each condition and task structure variation.

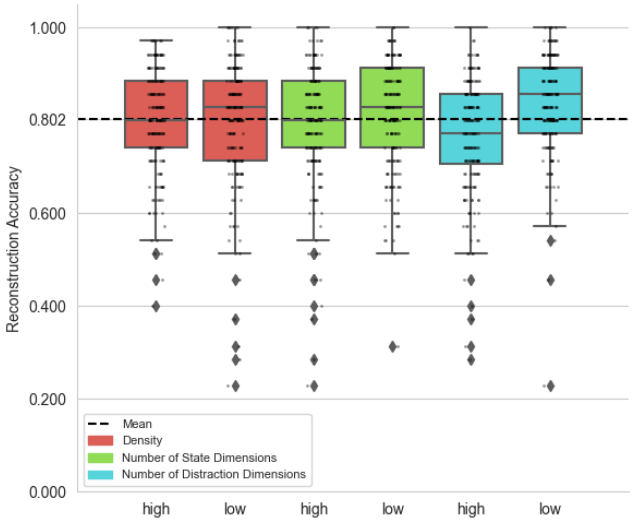


(a) IL-IL

(b) IRL-IL



(c) IL-IRL



(d) IRL-IRL

Figure A.3.: Distribution of the reconstruction accuracy for the Transfer-Condition. Each subplot illustrates a different model combination. The colors of the boxplots represent the variation types further distinguished between high and low. Black scatter points show each data point per boxplot. The dotted line in each of the four subplots shows the mean across all variations.

B. Appendix: Subject Experiment

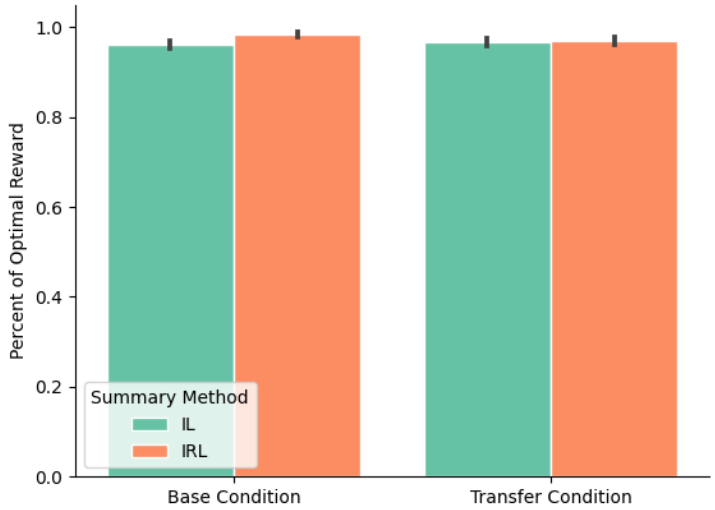
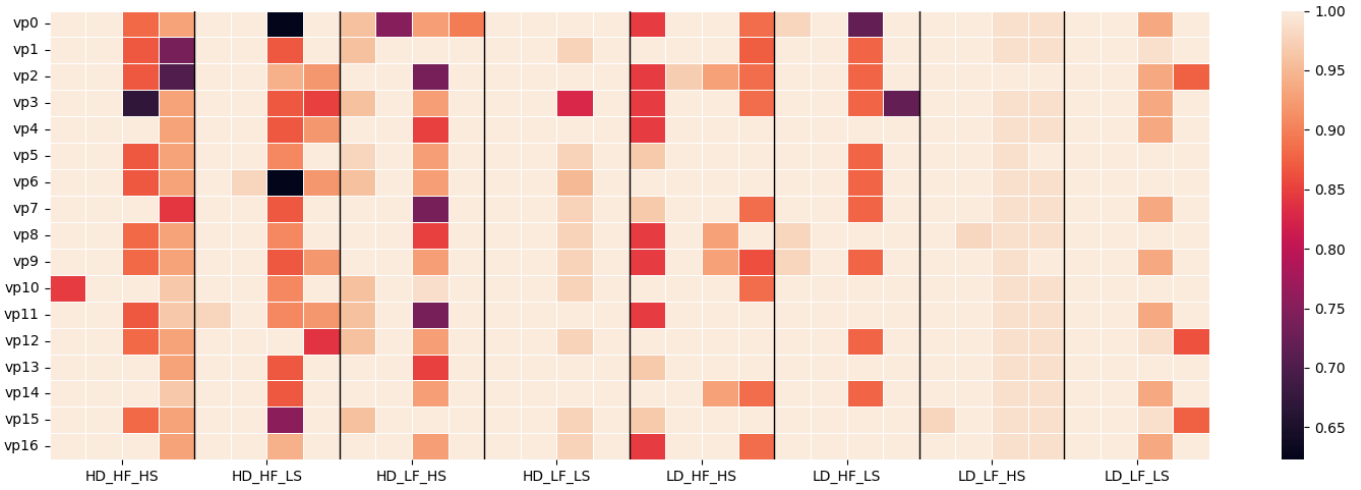
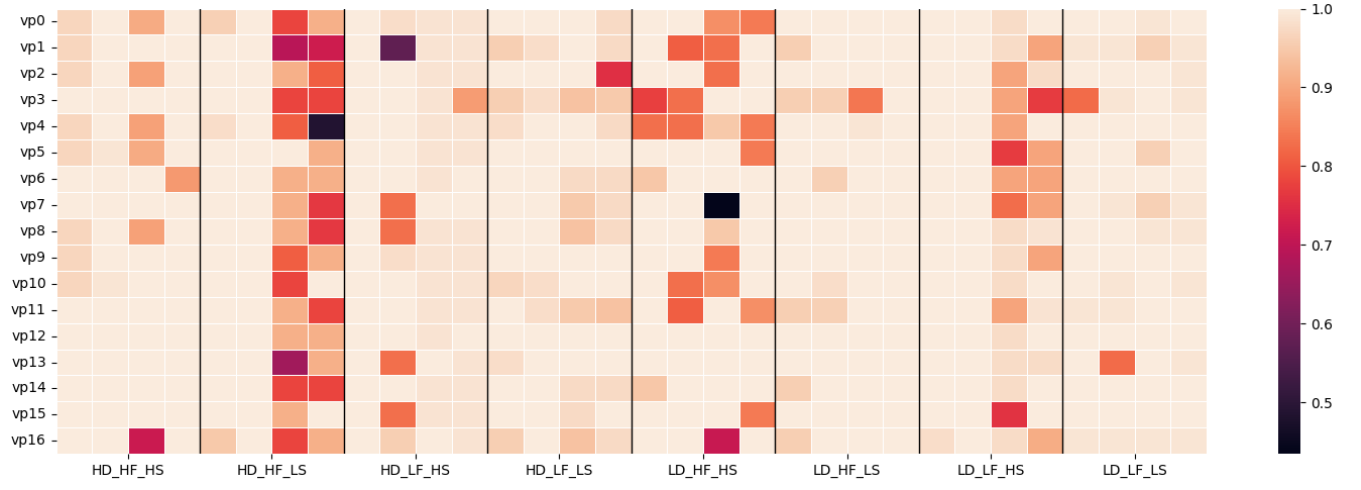


Figure B.1.: Mean performance for Base- and Transfer-Condition per summary method across all subjects. The error bars represent the 95% confidence interval.



(a) Base-Condition



(b) Transfer-Condition

Figure B.2.: Individual subject performance per trial. Each tile represents the performance per trial and subject. The x-axis labels represent different variation combinations, each denoted by abbreviations.